

CIS 5800

Machine Perception

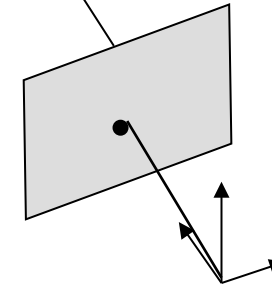
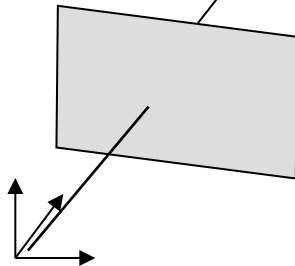
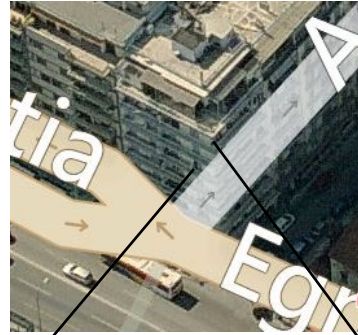
Instructor: Lingjie Liu

Lec 10 (Part 2): Feb 24, 2026

3D Motion from Two Views or Structure from Motion (SfM)

Input: Two Calibrated Views of the same 3D scene

Intrinsics known,
So we can always
stick to
“calibrated
coordinates” like
we used in P3P,
rather than pixel
coordinates.



Recap: Pixels → “Calibrated coordinates”

RGB images only provide pixel coordinates u, v for each vertex, not camera-centric 3D coordinates. Convert these to:

“Calibrated coordinates”: $\mathbf{p}_i \sim K^{-1}(u_i \quad v_i \quad 1)^T$

These are essentially image plane coordinates with principal point as origin, and focal length set to 1.

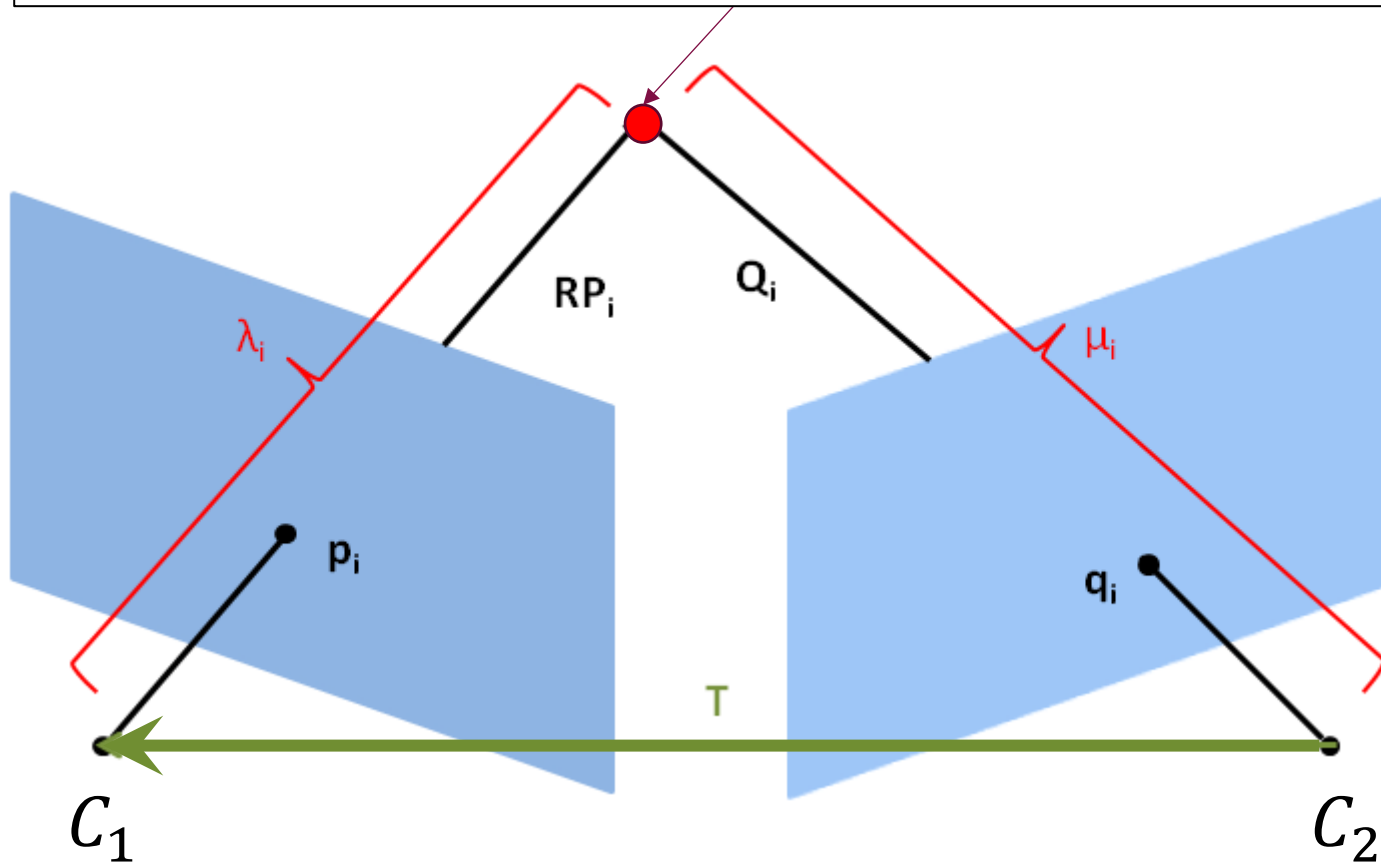
Euclidean interpretation: $\mathbf{p}_i$ is a vector in camera coordinates, originating from the camera center and pointing toward the 3D point corresponding to pixel coordinates (u_i, v_i) .

Recap: Calibrated coordinates + distance = Camera-centric 3D

- After writing as calibrated coordinates, camera-centric 3D coordinates is just one step away: finding the distance of the 3D point along that direction, i.e. “depth”
- To reduce P3P to 3D->3D as we had planned, we will need to find those depths. Indeed, that is the first step of P3P.

Input: Two Calibrated Views of the Same 3D Scene

$[\lambda p]$ in C_1 camera coords or $[\mu q]$ in C_2 camera coords



$$[X]_{C_2} = R[X]_{C_1} + T$$

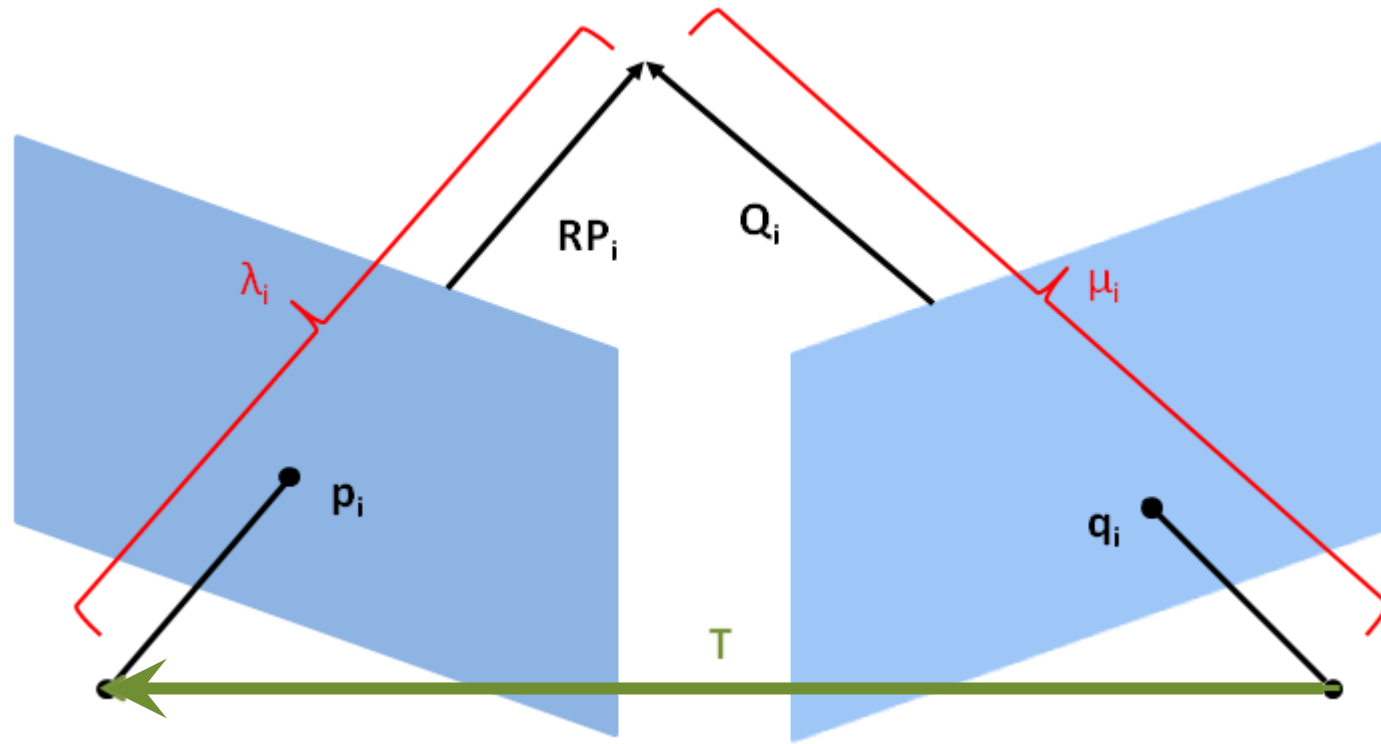
Now, any 3D coordinates in C_1 frame:
 $[X]_{C_1} \rightarrow [R[X]_{C_1} + T]_{C_2}$

So $[\lambda p]_{C_1} \rightarrow [R(\lambda p) + T]_{C_2} = [\mu q]_{C_2}$

Or now that we are in the same coordinate system:

$$R(\lambda p) + T = \mu q$$

Two Calibrated Views of the Same 3D Scene



$$R(\lambda p) + T = \mu q$$

Given 2D correspondences (p, q)

Find motion R, T and depths λ, μ .

PnP vs. 2-View Structure from Motion (SfM)

PnP

1. Gn. “world frame” points P_i and corresponding calibrated coordinates $K^{-1}x_i$
2. Set camera frame 3D coordinates (with scale/depth ambiguity) to $\lambda_i K^{-1}x_i$
3. Then solve (for R, T, λ):

$$\lambda_i K^{-1}x_i = R P_i + T \quad \forall i$$

Where R, T denotes transformation between camera and world.

Structure from Motion

1. No “world frame”. Instead, just calibrated image plane coordinates $p_i = K_1^{-1}x_{i1}$ and $q_i = K_2^{-1}x_{i2}$ **of the same 3D point P_i .**
2. So corresponding camera frame 3D coordinates in the two frames are: $\lambda_i p_i$ and $\mu_i q_i$.
3. Now solve for “motion between cameras” R, T and the scales λ_i, μ_i (which permit getting 3D coordinates of P_i in either camera frame)

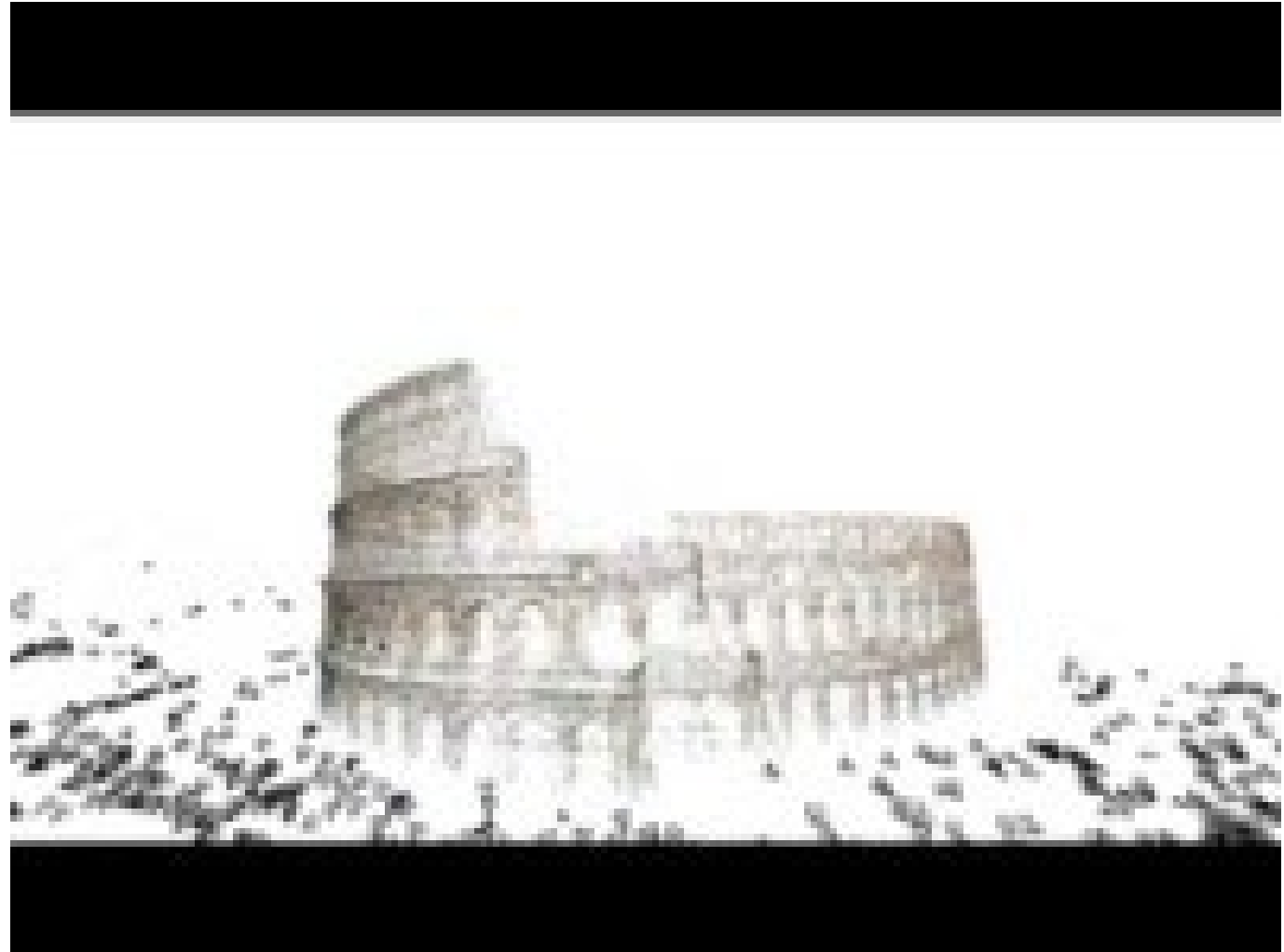
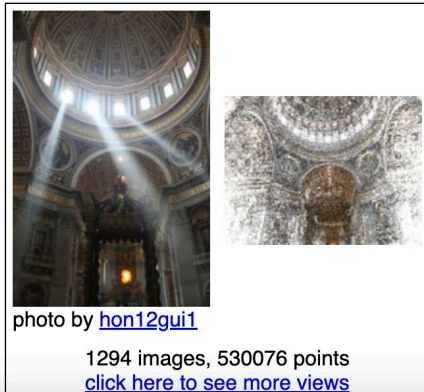
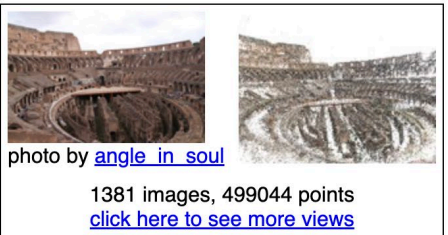
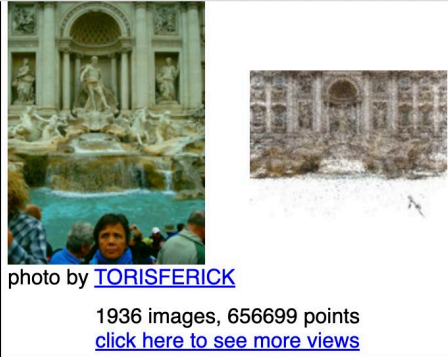
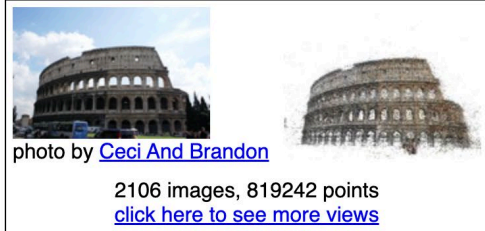
SfM, SLAM, Visual Odometry

- **“SfM” (structure from motion):** graphics and computer vision folks interested mainly in building 3D models of scenes/objects. Images could be unordered, from multiple different cameras taking pictures of the same 3D scene/object. Usually performed offline.
- **“VO” (visual odometry):** mainly interested in localizing and tracking the robot alone, i.e. how far has it traveled in what direction, what speed etc. over short time window. Images usually frames of a video captured from a moving camera in a static / close-to-static world. Often performed online, as images stream in. Usually a single pre-calibrated camera.
- **“SLAM” (simultaneous localization and mapping):** Jointly estimating 3D models of scenes and localizing the robot (camera) w.r.t. that scene. Images similar to VO, also streaming, but emphasis on long-term map consistency. VO++. We'll see an example of a SLAM system later.

Methods largely the same, with differences in emphases depending on which output (depths, transformations, or both) matter most.

SfM Application: Building Rome in a Day

SfM techniques applied to Flickr image collections!

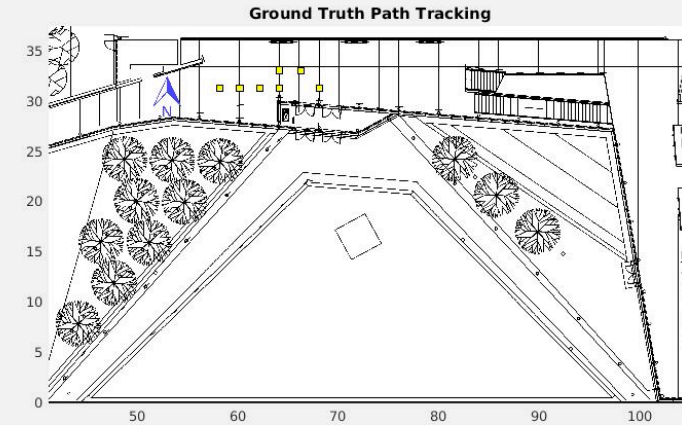
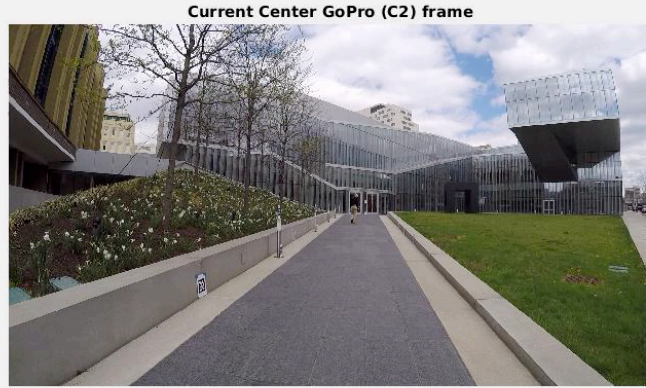


SfM Application: 3D Presidential Portrait



Really MVS
rather than SfM,
since the camera
locations are
fully known.

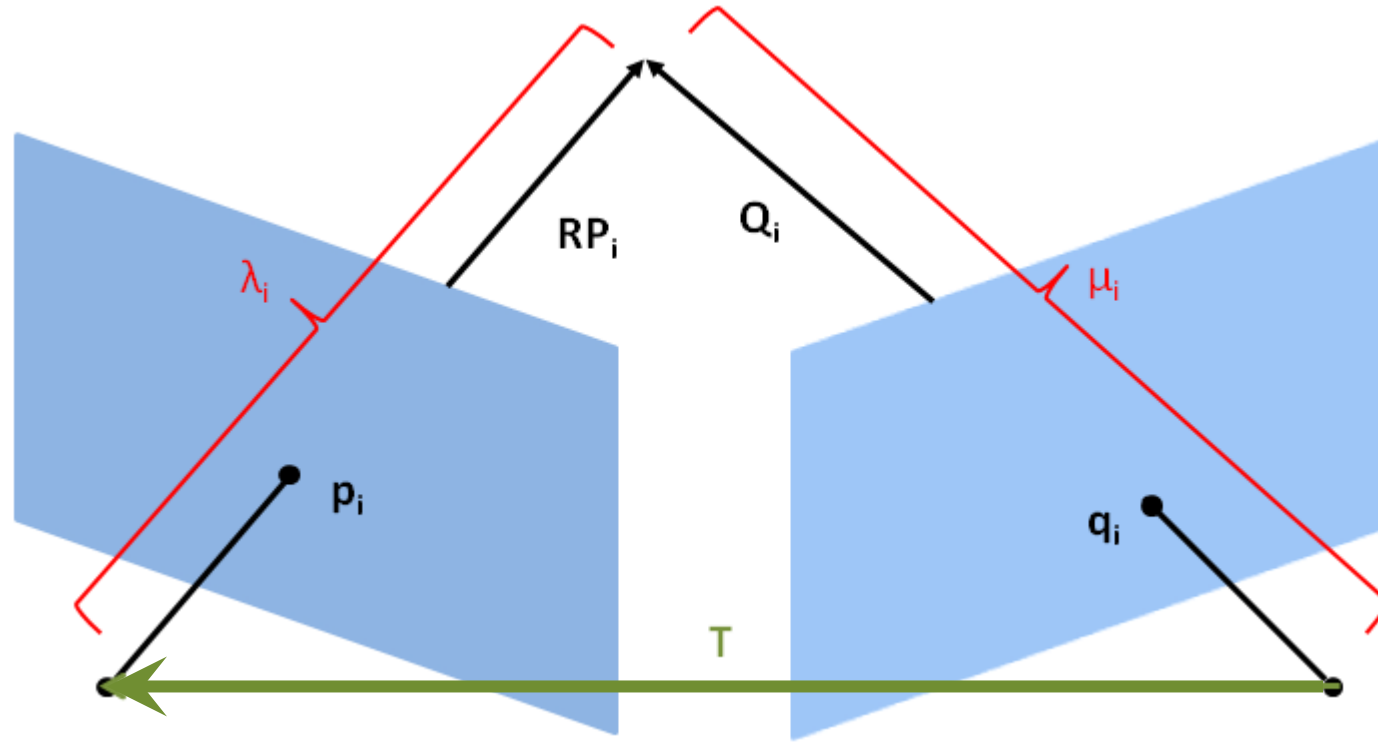
SfM is the main ingredient of Visual Odometry



SLAM for Driving in DARPA Grand Challenge (circa 2007)



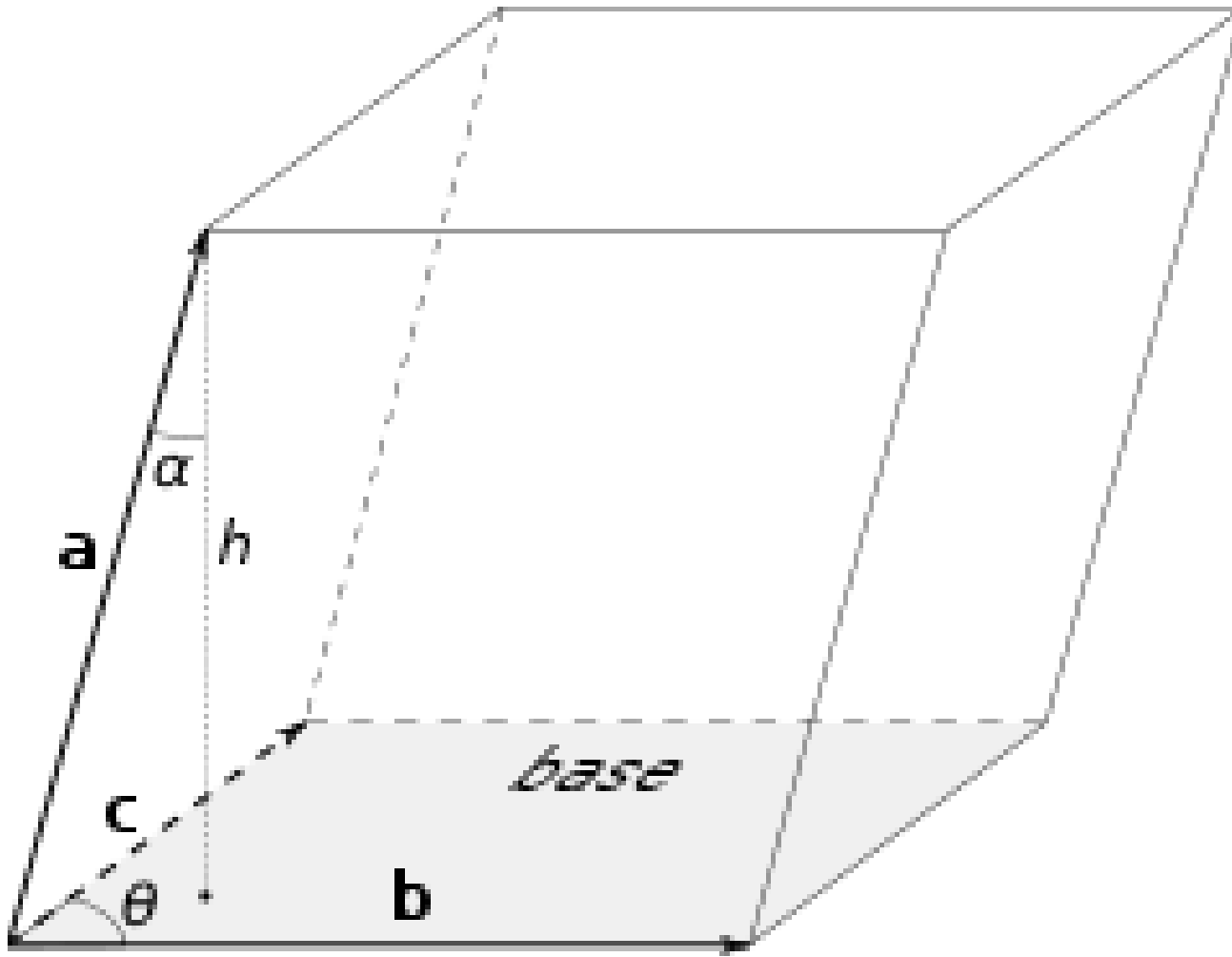
“Epipolar Constraints” Between Two Views of a Scene



We can eliminate the depths from $R(\lambda p) + T = \mu q$ and obtain the epipolar constraint:

$$q_i^T (T \times R p_i) = 0$$

Review: Mixed Product = Volume of Parallelepiped



$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$$

In our setting, q, Rp, T are edges of a triangle because

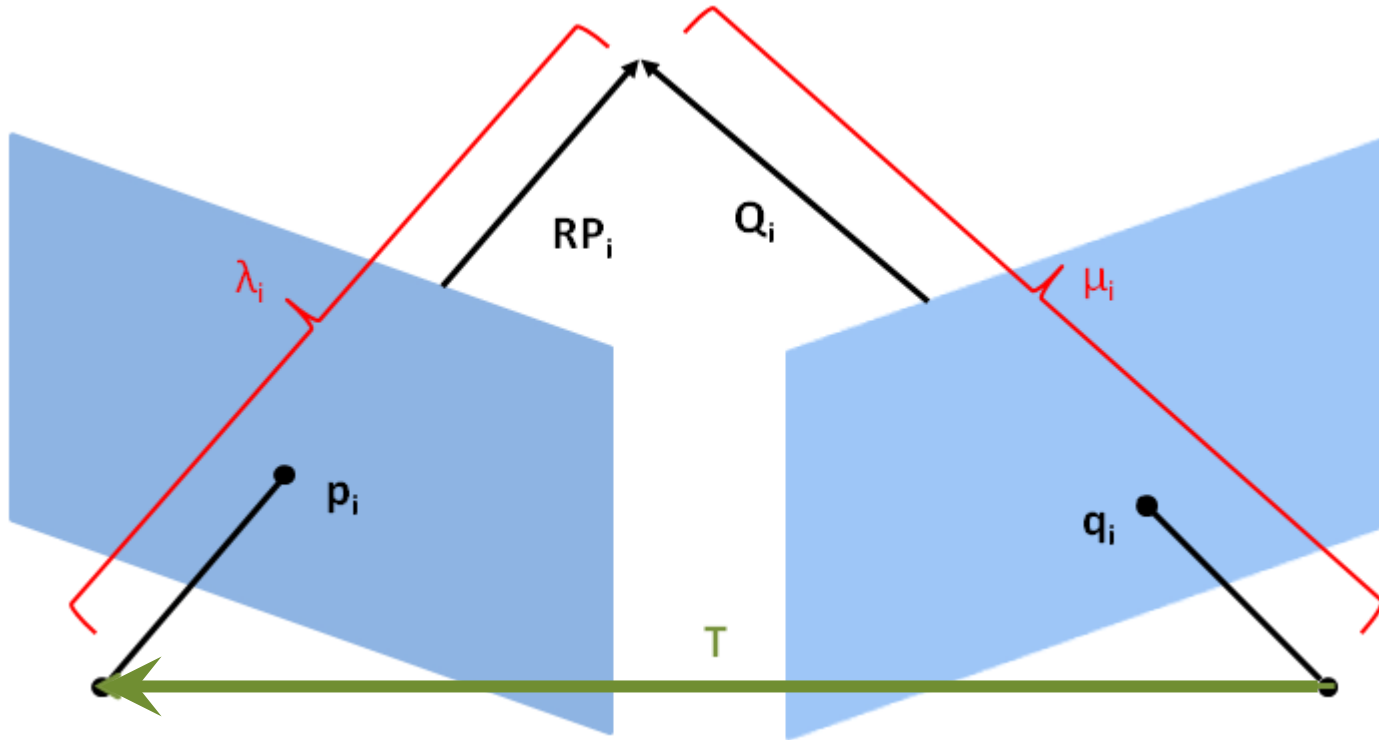
$$\lambda R\mathbf{p} + \mathbf{T} = \mu \mathbf{q}$$

(which is the triangle rule for vectors).

And triangles are planar!

Fun fact: this is also the determinant of a matrix with columns $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

Geometric Meaning of Epipolar Constraint



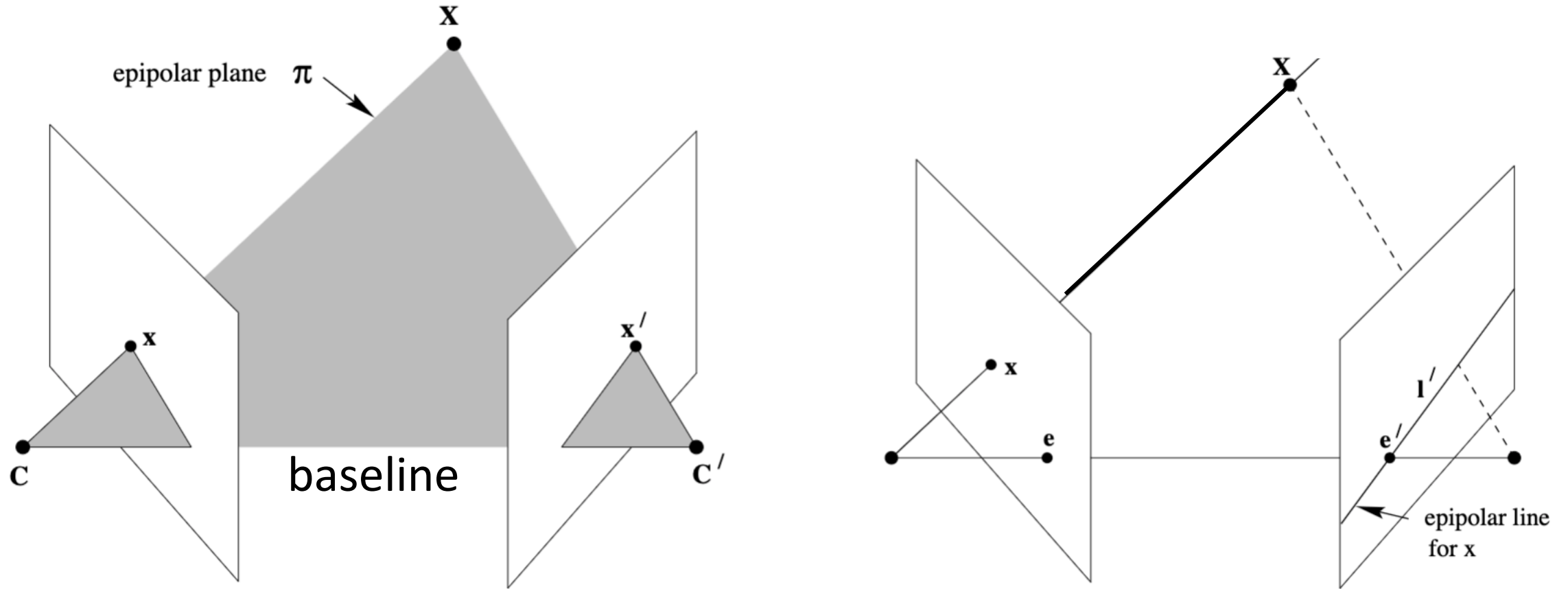
The two rays q and Rp intersect in space if and only if they are coplanar with the translation vector T . Three vectors are coplanar if their mixed product vanishes:

$$q^T(T \times Rp) = 0$$

Epipolar constraint =
“image rays from the world point to the two cameras lies on the same plane as the baseline (translation vector) connecting the camera centers.”

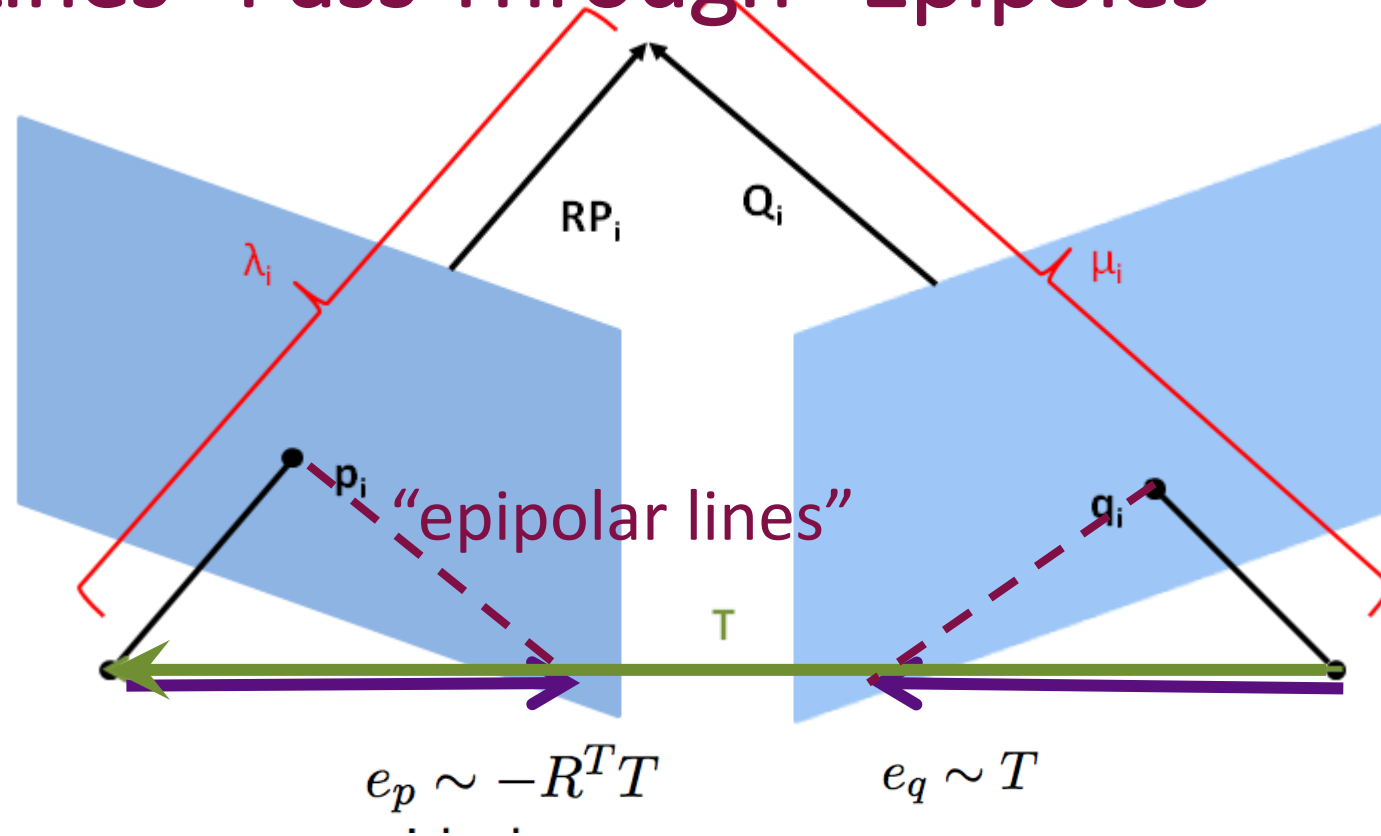
Again, these are just the edges of a triangle with the 2 camera centers and the world point as vertices, so naturally coplanar!

Epipolar “Planes” and “Lines”



Epipolar planes are planes containing the baseline.
Any 3D point induces a corresponding epipolar plane.
Intersection of this plane with an image plane = epipolar line for that 3D point.

“Epipolar Lines” Pass Through “Epipoles”



$e_p \sim -R^T T$ and $e_q \sim T$ are the “epipoles” = images of the other camera center on each plane = intersections of baseline T with the two planes = VP of the translation direction in each plane.

All epipolar lines in each image plane pass through its epipole.

Collecting Epipolar Lines into Epipolar “Pencils”

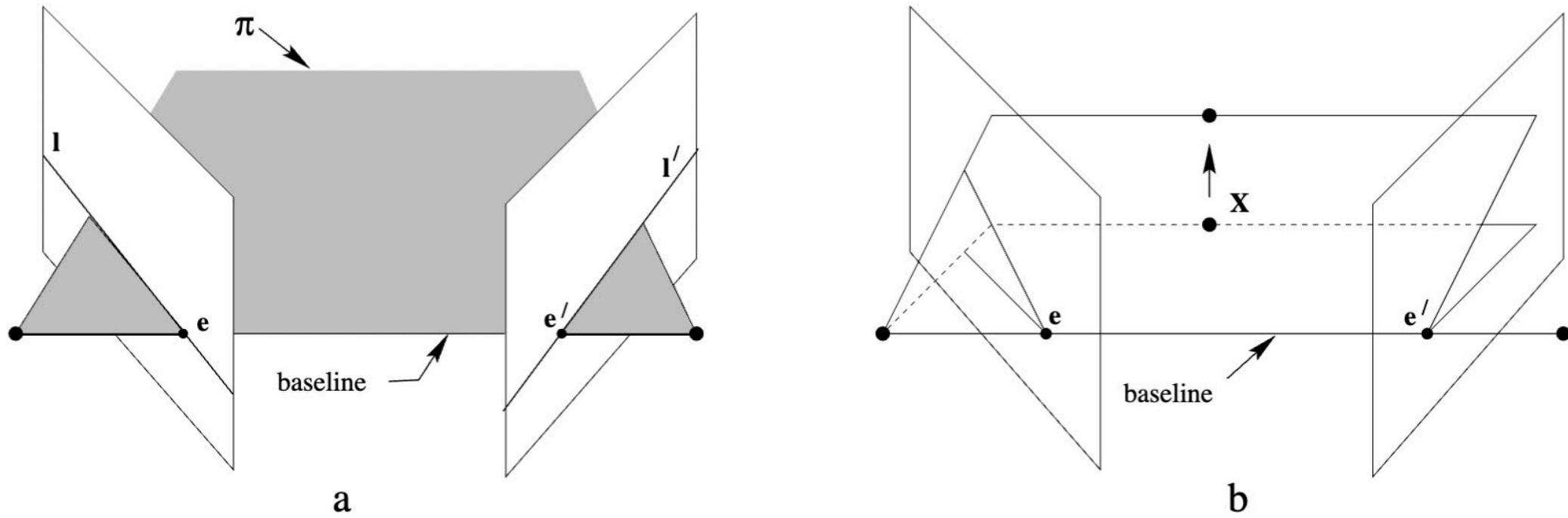
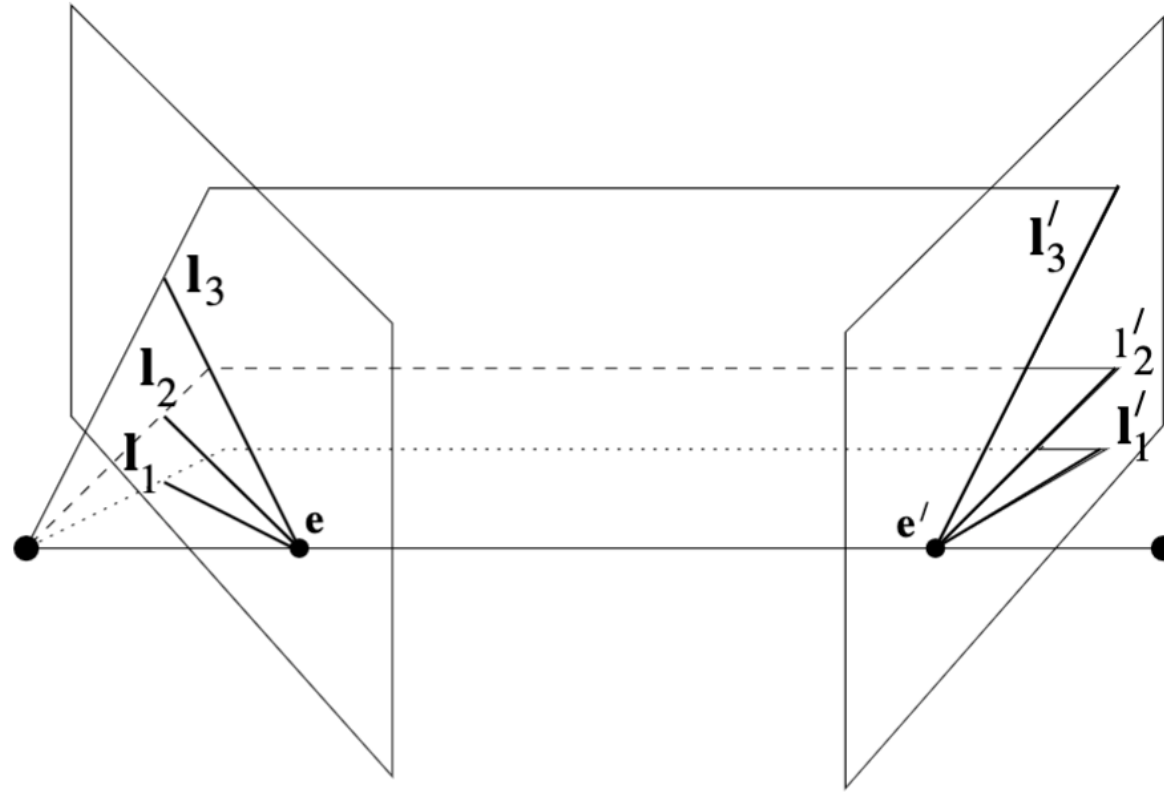


Fig. 9.2. **Epipolar geometry.** (a) The camera baseline intersects each image plane at the epipoles e and e' . Any plane π containing the baseline is an epipolar plane, and intersects the image planes in corresponding epipolar lines l and l' . (b) As the position of the 3D point X varies, the epipolar planes “rotate” about the baseline. This family of planes is known as an epipolar pencil. All epipolar lines intersect at the epipole.

Epipolar “Pencils”



The epipolar lines in each image plane form a “pencil” whose tip is the epipole i.e. image of the other camera center / intersection of baseline with image plane. (Because all epipolar planes contain the baseline, after all)

Epipolar Pencils of Frontoparallel Cameras

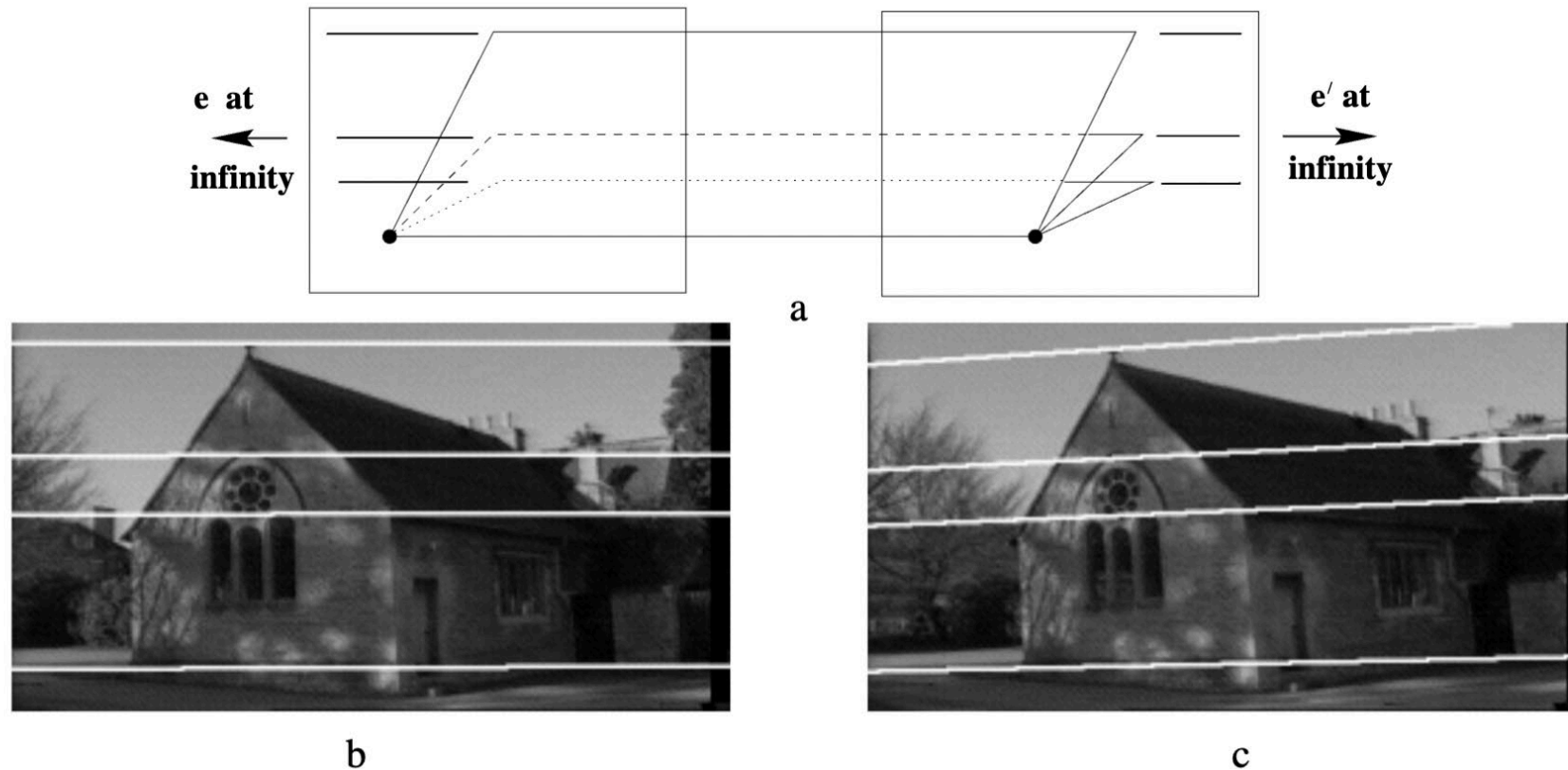
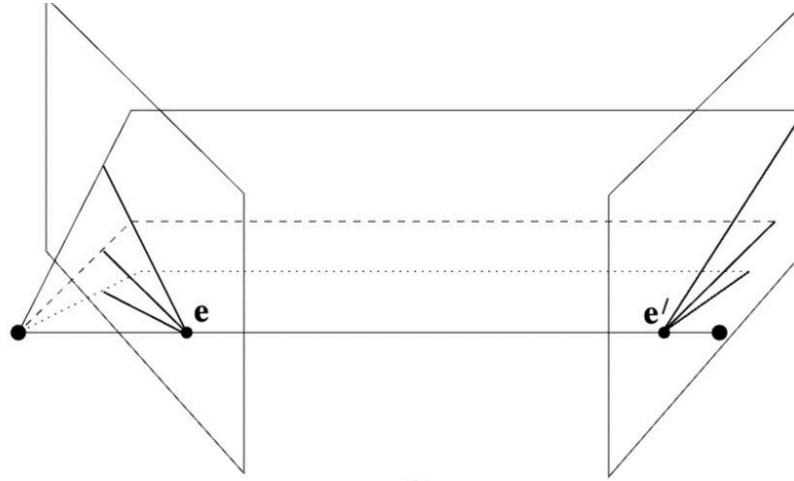


Fig. 9.4. **Motion parallel to the image plane.** *In the case of a special motion where the translation is parallel to the image plane, and the rotation axis is perpendicular to the image plane, the intersection of the baseline with the image plane is at infinity. Consequently the epipoles are at infinity, and epipolar lines are parallel. (a) Epipolar geometry for motion parallel to the image plane. (b) and (c) a pair of images for which the motion between views is (approximately) a translation parallel to the x -axis, with no rotation. Four corresponding epipolar lines are superimposed in white. Note that corresponding points lie on corresponding epipolar lines.*

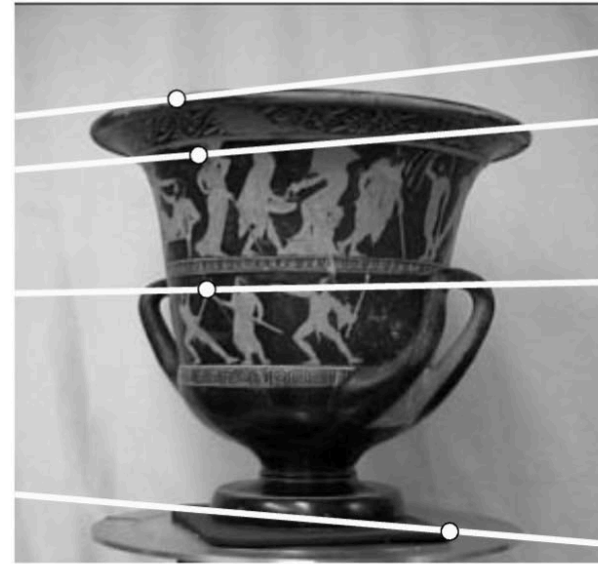
Epipolar Pencils visually identify camera orientations



a



b



c

Epipolar Pencils visually identify camera orientations

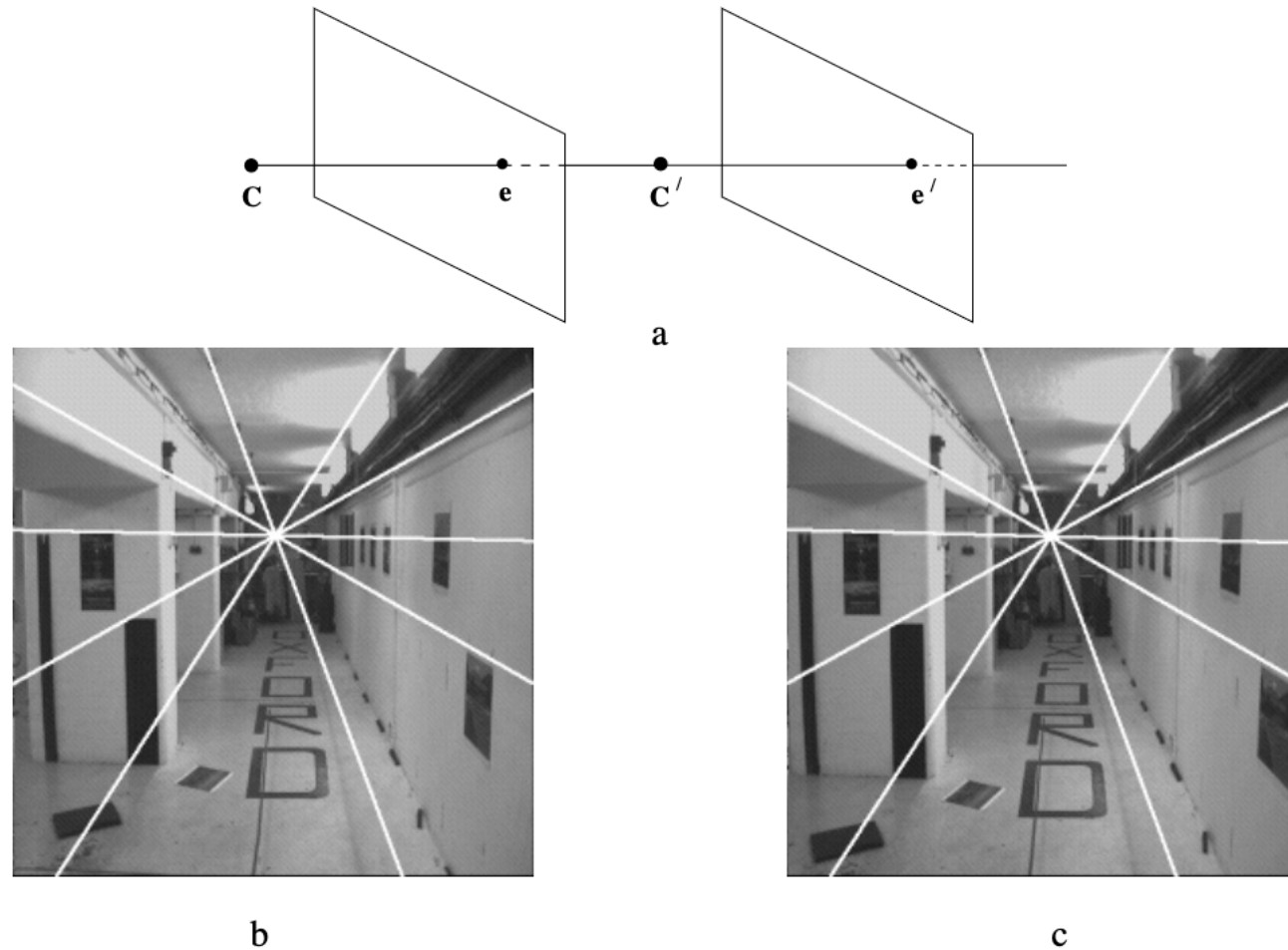


Fig. 9.8. **Pure translational motion.** (a) under the motion the epipole is a fixed point, i.e. has the same coordinates in both images, and points appear to move along lines radiating from the epipole. The epipole in this case is termed the Focus of Expansion (FOE). (b) and (c) the same epipolar lines are overlaid in both cases. Note the motion of the posters on the wall which slide along the epipolar line.

CIS 5800

Machine Perception

Instructor: Lingjie Liu

Lec 11: Feb 26, 2026

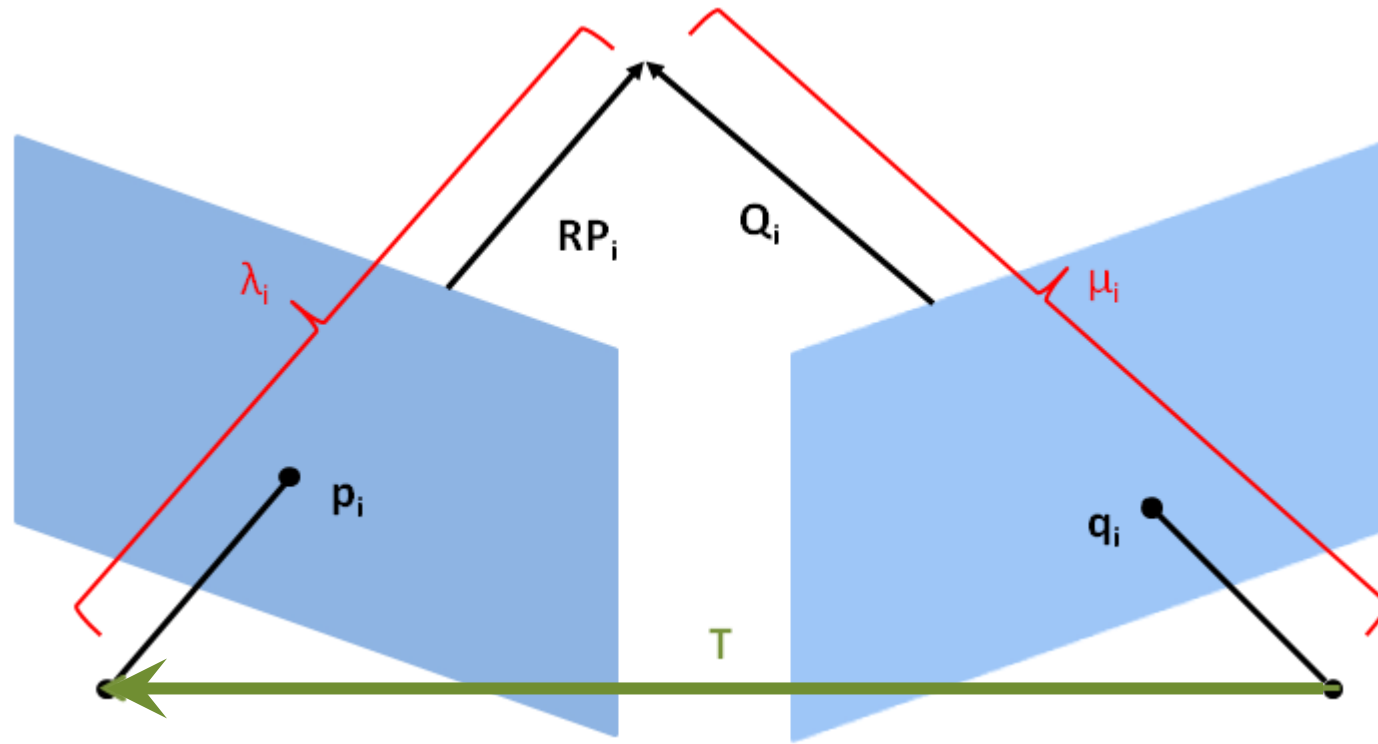
Administrivia

- Midterm exam coming up
 - Date reminder: Wednesday March 26 (class hours, we'll 3:30pm exactly)
 - Syllabus: All material covered in class until Tue March 17.
 - Review questions pdf will be released next week
 - If you are unable to attend the midterm exam in person on March 26, please complete the form by March 17:
<https://docs.google.com/forms/d/e/1FAIpQLSfzXSMvI4DqEdeos-WDEMFbgn7I1BJJ-kpSGLVkolaJZgJsWg/viewform?usp=dialog>
 - Also, you need to contact the Weingarten Office for academic accommodations and send me the paperwork or approval from the Weingarten Office.

Administrivia

- HW 2 is released and the due date is March 17.

Recap: Two Calibrated Views of the Same 3D Scene

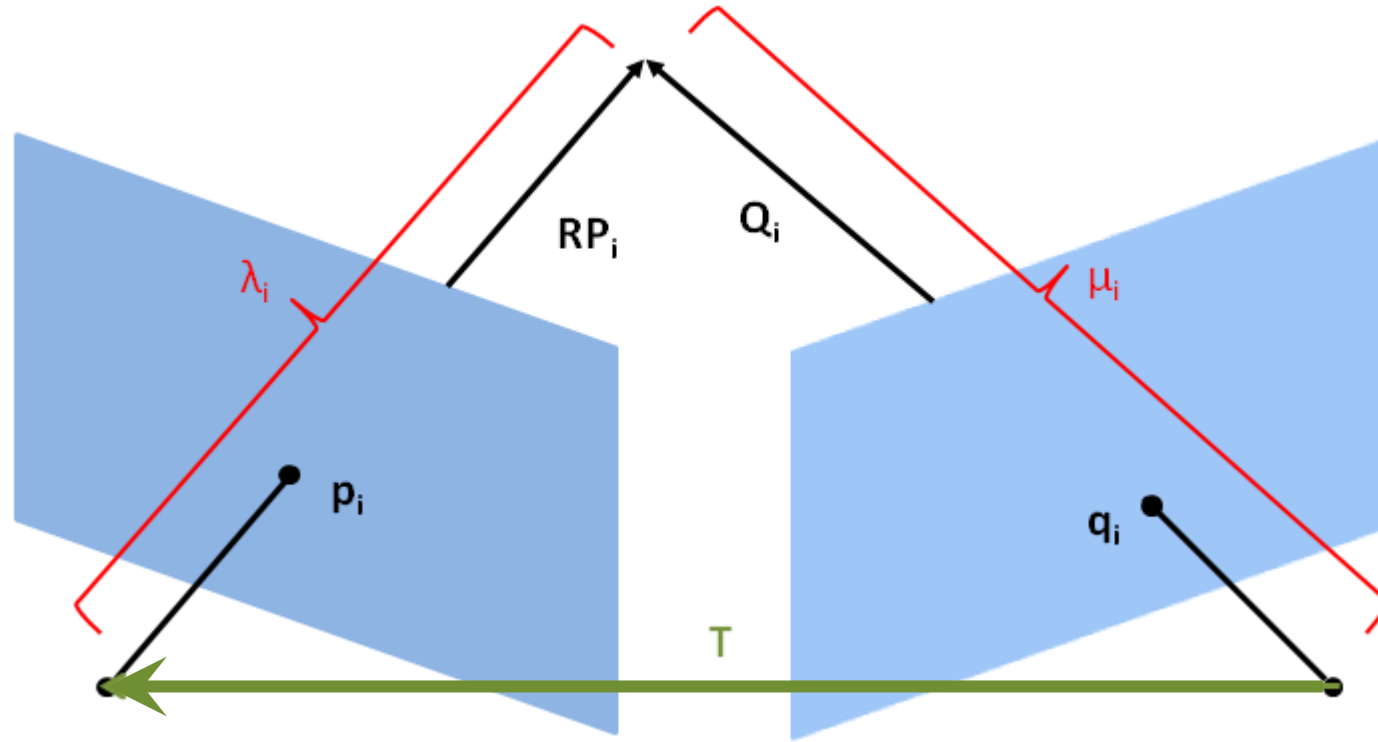


$$R(\lambda p) + T = \mu q$$

Given 2D correspondences (p, q)

Find motion R, T and depths λ, μ .

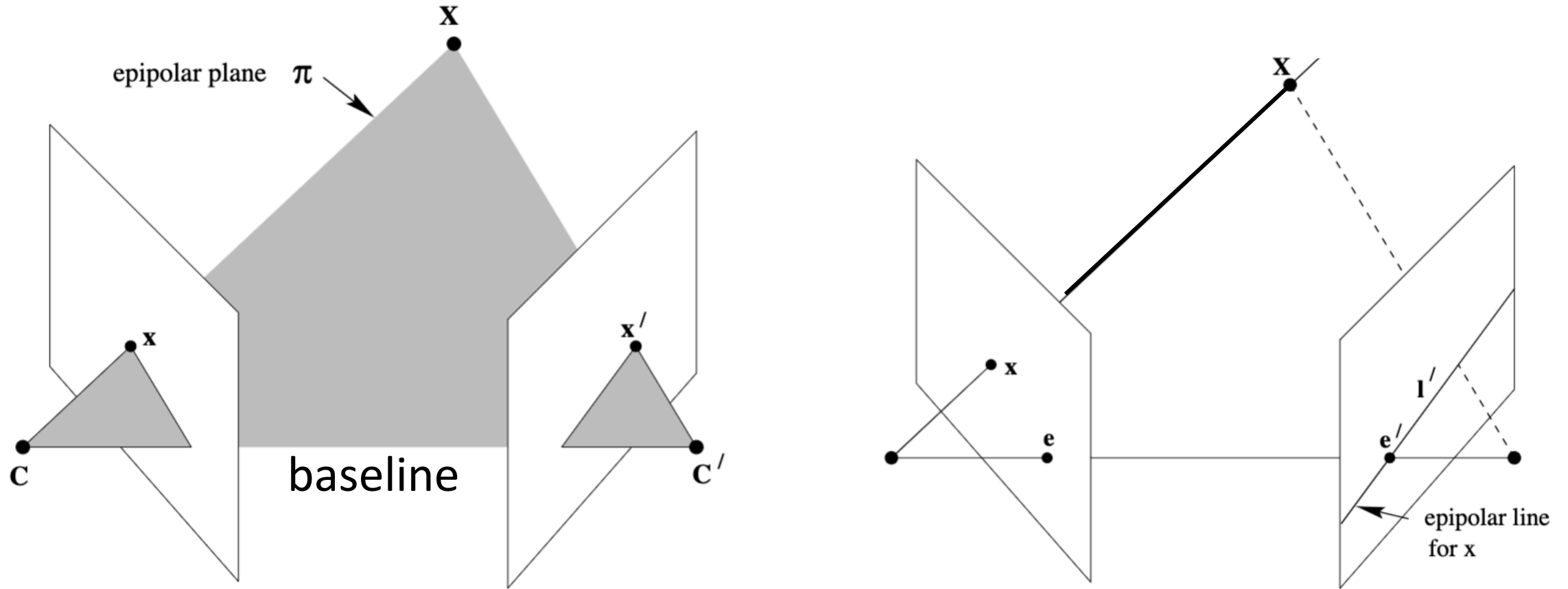
Recap: “Epipolar Constraints” Between Two Views of a Scene



We can eliminate the depths from $R(\lambda p) + T = \mu q$ and obtain the epipolar constraint:

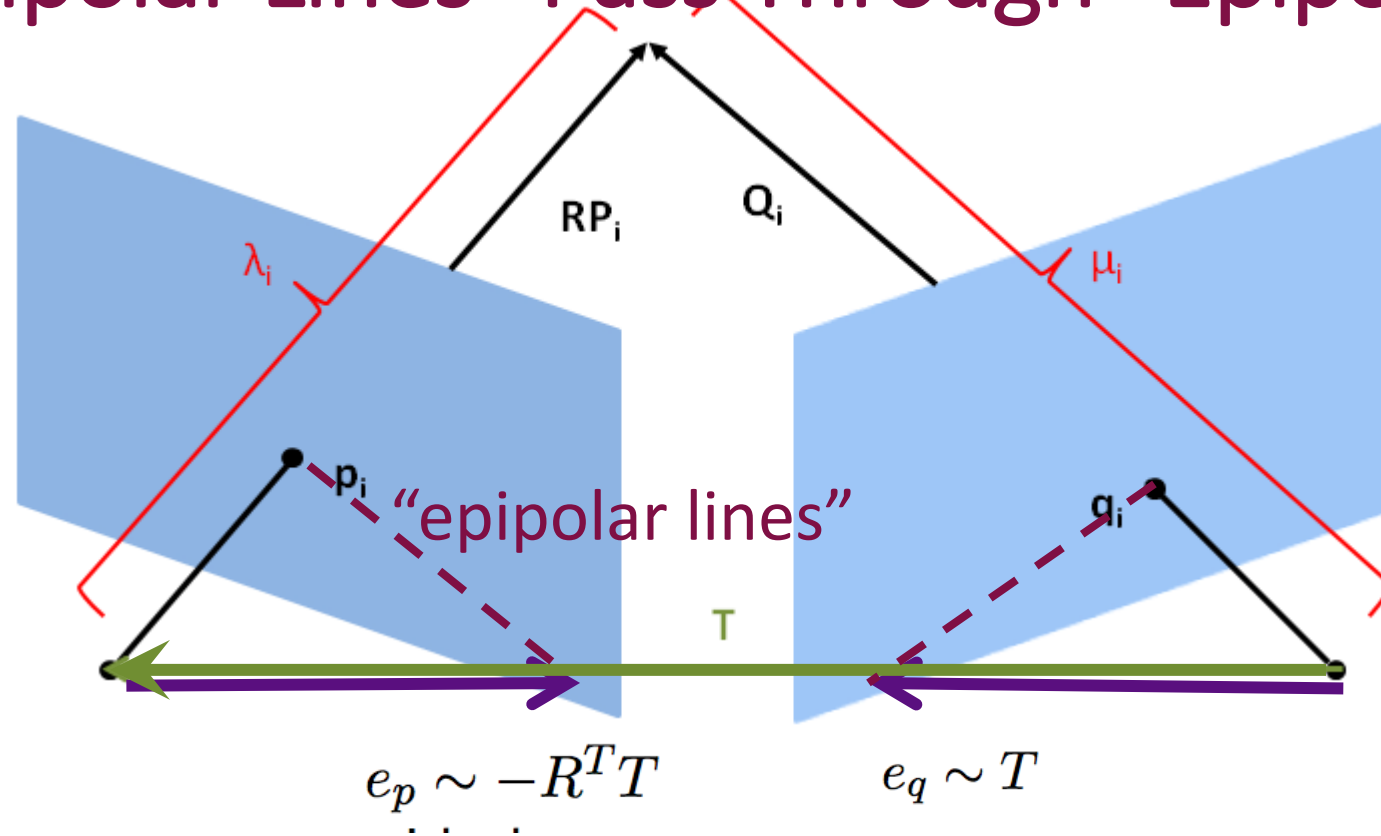
$$q_i^T (T \times R p_i) = 0$$

Recap: Epipolar “Planes” and “Lines”



Epipolar planes are planes containing the baseline.
Any 3D point induces a corresponding epipolar plane.
Intersection of this plane with an image plane = epipolar line for that 3D point.

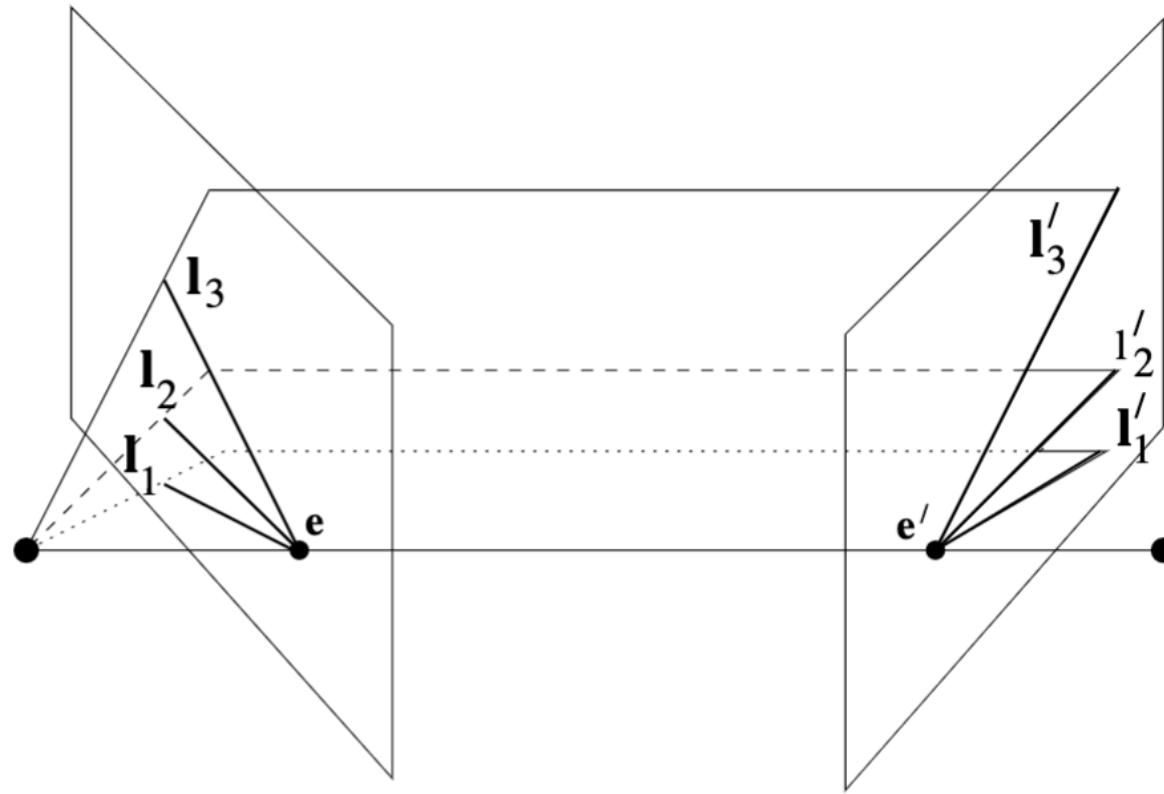
Recap: “Epipolar Lines” Pass Through “Epipoles”



$e_p \sim -R^T T$ and $e_q \sim T$ are the “epipoles” = images of the other camera center on each plane = intersections of baseline T with the two planes = VP of the translation direction in each plane.

All epipolar lines in each image plane pass through its epipole.

Recap: Epipolar “Pencils”



The epipolar lines in each image plane form a “pencil” whose tip is the epipole i.e. image of the other camera center / intersection of baseline with image plane. (Because all epipolar planes contain the baseline, after all)

Recap: Epipolar Constraints

$$\mathbf{q}_i^T (\mathbf{T} \times \mathbf{R} \mathbf{p}_i) = 0$$

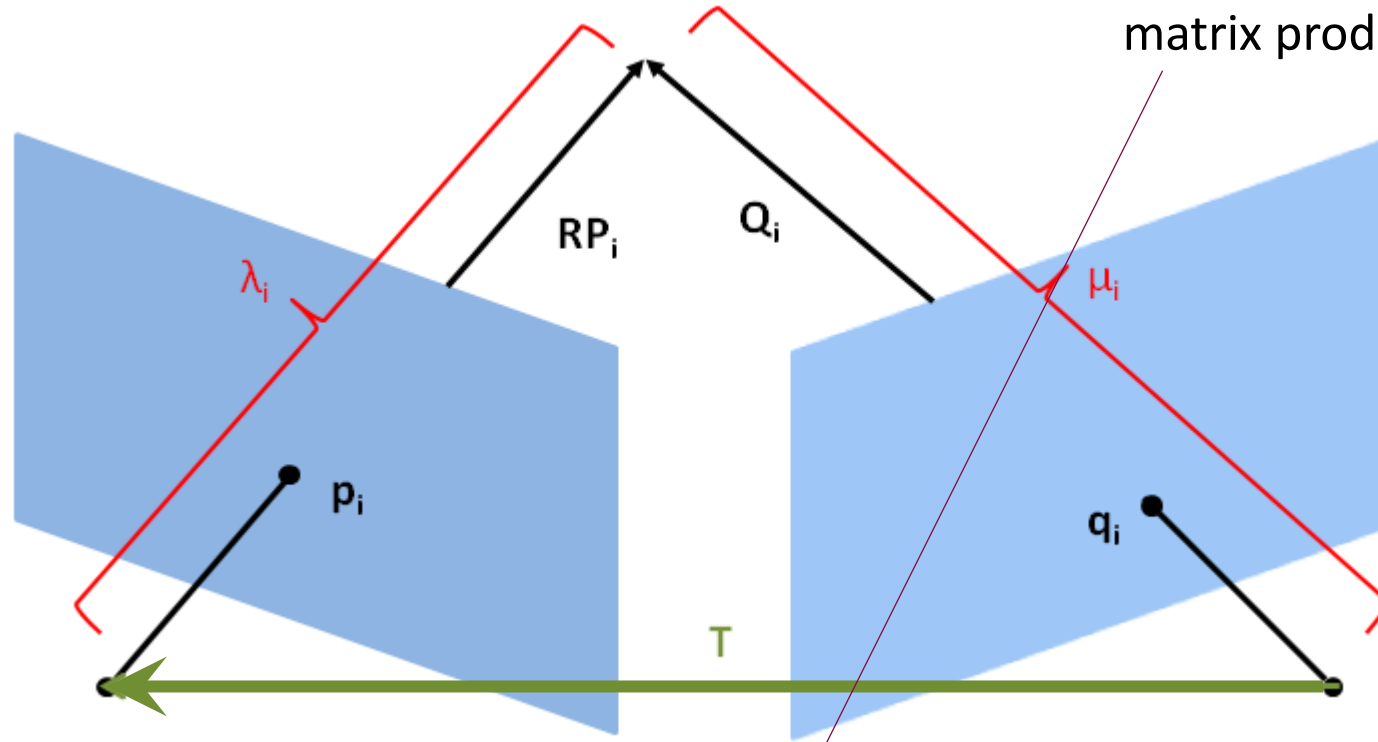
Gives us an equation in unknowns: $\mathbf{R}, \mathbf{T} \Rightarrow$ route to an SfM solution?

Q: If this were linear in $\mathbf{R}, \mathbf{T}$, we could solve it with enough $(\mathbf{p}_i, \mathbf{q}_i)$ pairs i.e. 2D- $\rightarrow$ 2D correspondences. Is it linear though?

A: No, but through a change of variables, we will soon make the equation linear in some new unknowns.

Also note: we are not only interested in $\mathbf{R}, \mathbf{T}$ (the “motion”). We also want to solve for the depths λ_i, μ_i afterwards (the “structure”).

Epipolar Constraints



Want to solve for R, T .
Can we make this look more like
standard linear equations containing
matrix products etc.?

We can eliminate the depths from $R(\lambda p) + T = \mu q$ and obtain the epipolar constraint:

$$q^T (T \times Rp) = 0$$

Review: Cross-products through skew-symmetric matrices

The vector cross product also can be expressed as the product of a **skew-symmetric matrix** and a vector:

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Sometimes written as $\hat{a}$. Remember, this is now a 3×3 matrix, while original a, b were 3×1 vectors.

The Essential Matrix E

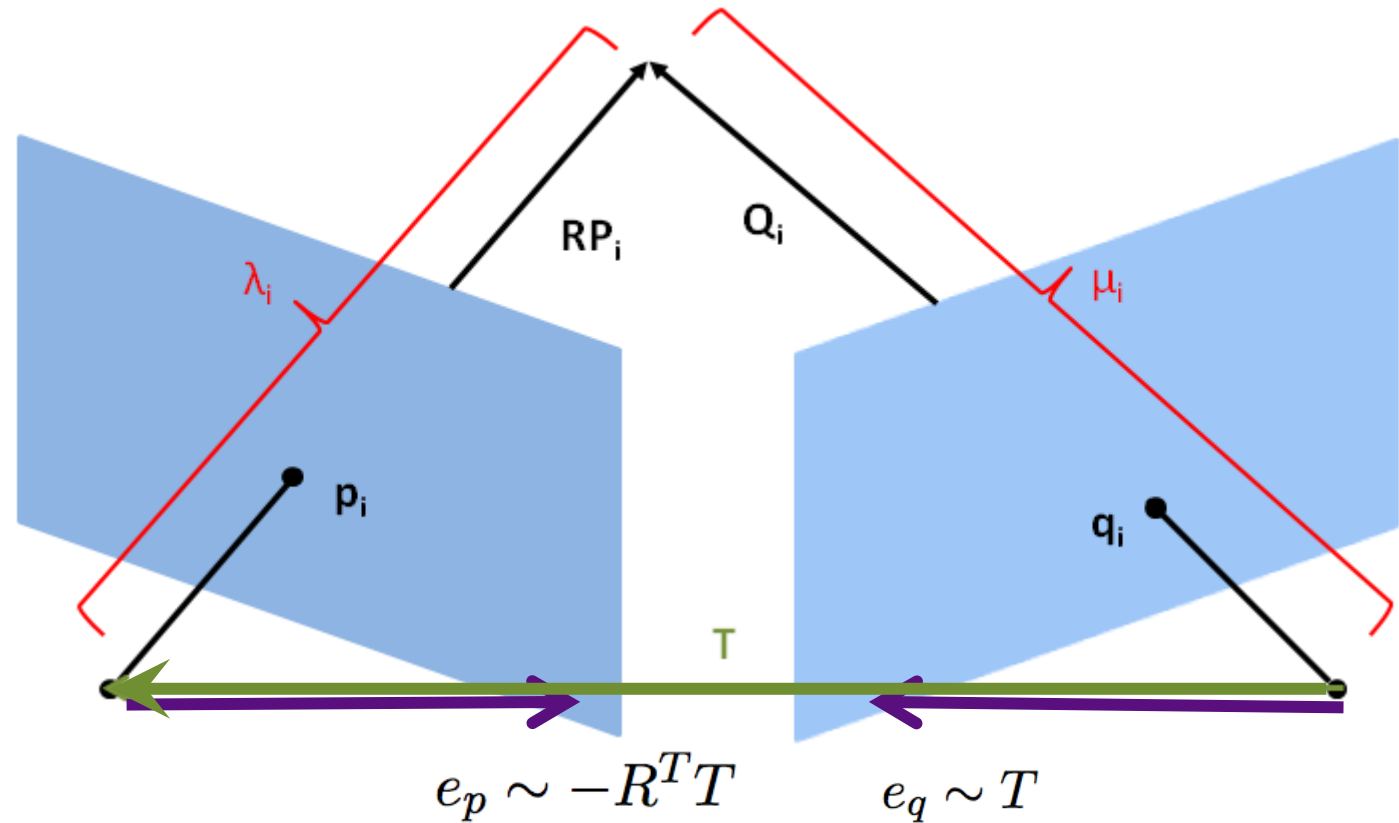
We had: $\mathbf{q}_i^T (\mathbf{T} \times \mathbf{R} \mathbf{p}_i) = 0$

$$\Rightarrow \mathbf{q}_i^T (\hat{\mathbf{T}} \mathbf{R}) \mathbf{p}_i = 0$$

Renaming $\mathbf{E} = (\hat{\mathbf{T}} \mathbf{R})$:

$$\mathbf{q}_i^T \mathbf{E} \mathbf{p}_i = 0$$

↓
“Essential matrix”



Now linear in the new unknowns $\mathbf{E}_{3 \times 3}$! But will need to recover $\mathbf{T}_{3 \times 1}$, $\mathbf{R}_{3 \times 3}$ later.

Is the epipolar constraint really linear in E ?

$q_i^T E_{3 \times 3} p_i = 0$ is a single equation that is linear in the elements of E
Can write this out explicitly as below.

If

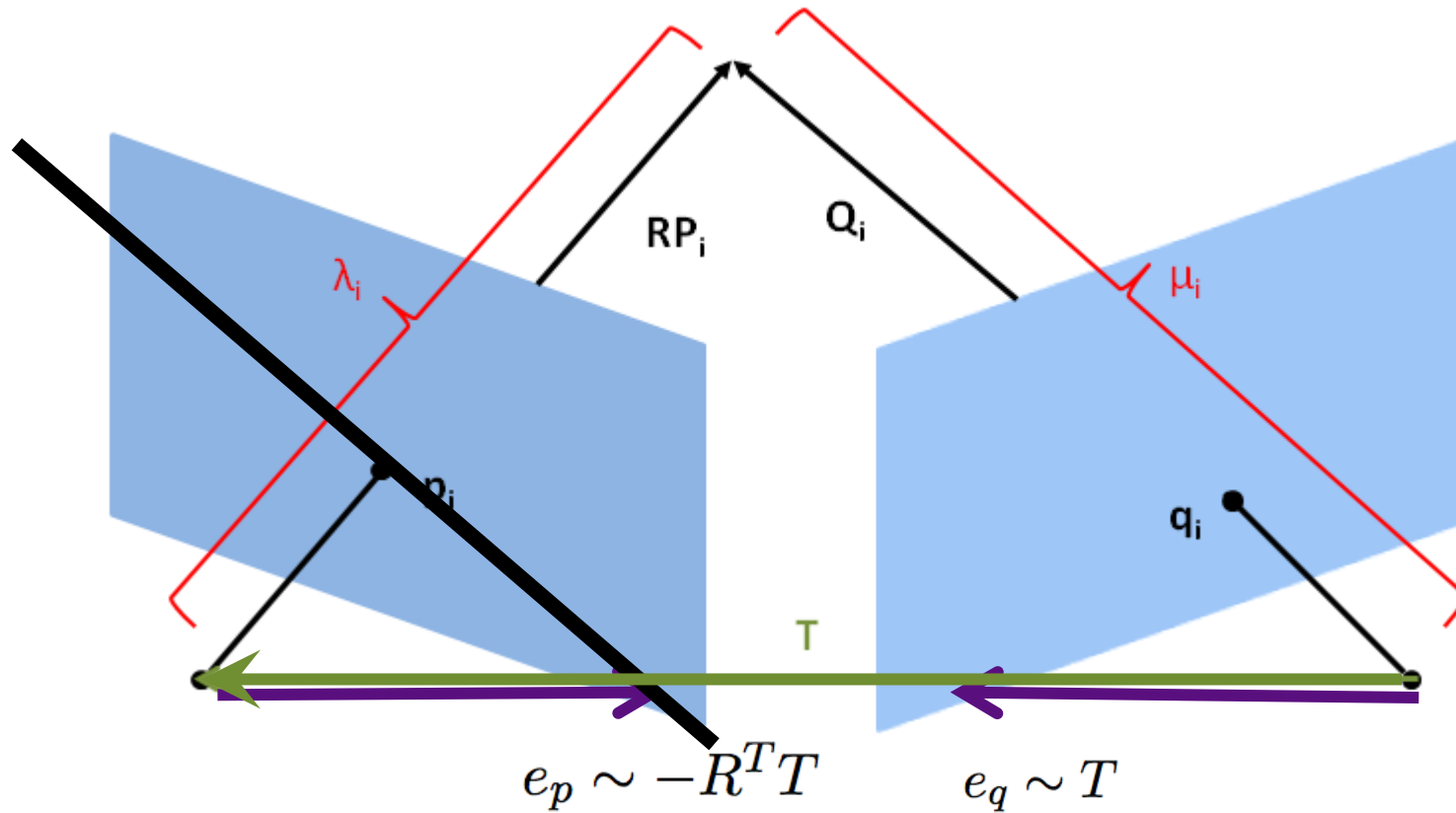
$$E = (e_1 \quad e_2 \quad e_3)$$

then epipolar constraint can be rewritten as

$$\begin{aligned} q^T (e_1 \quad e_2 \quad e_3) \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} &= q^T (p_x e_1 + p_y e_2 + p_z e_3) \\ &= (p_x q^T \quad p_y q^T \quad p_z q^T) \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = 0 \end{aligned}$$

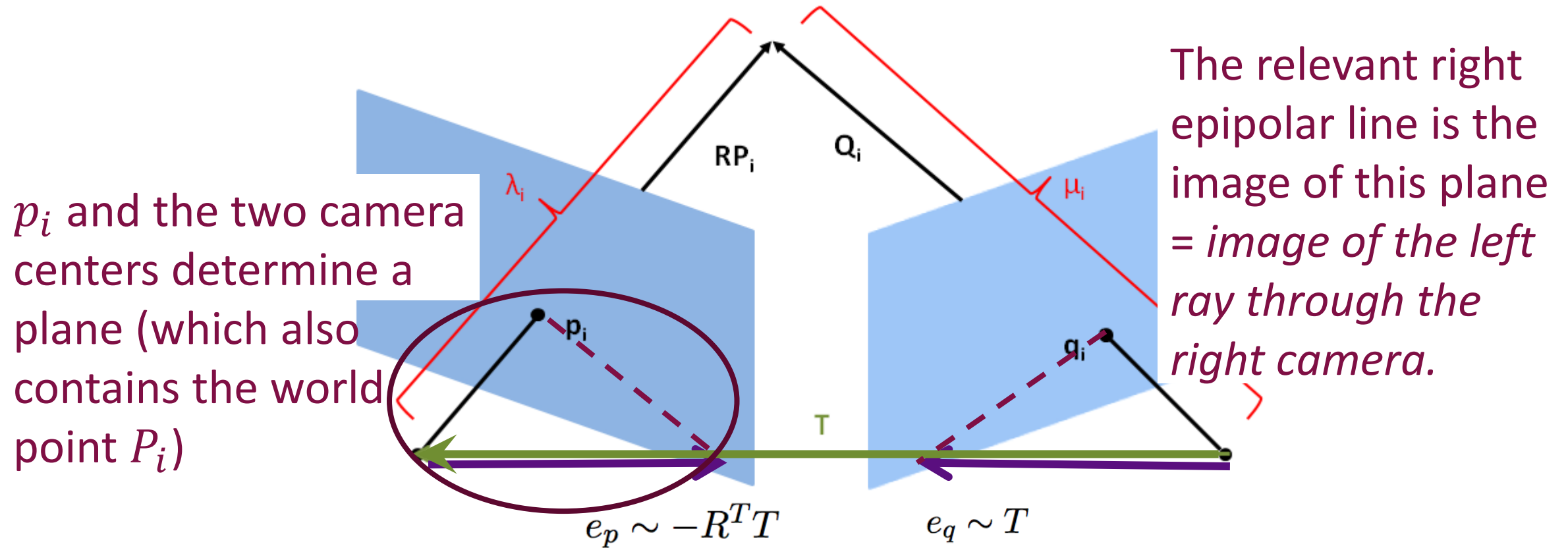
This equation is linear

Epipolar Lines in Essential Matrix Notation



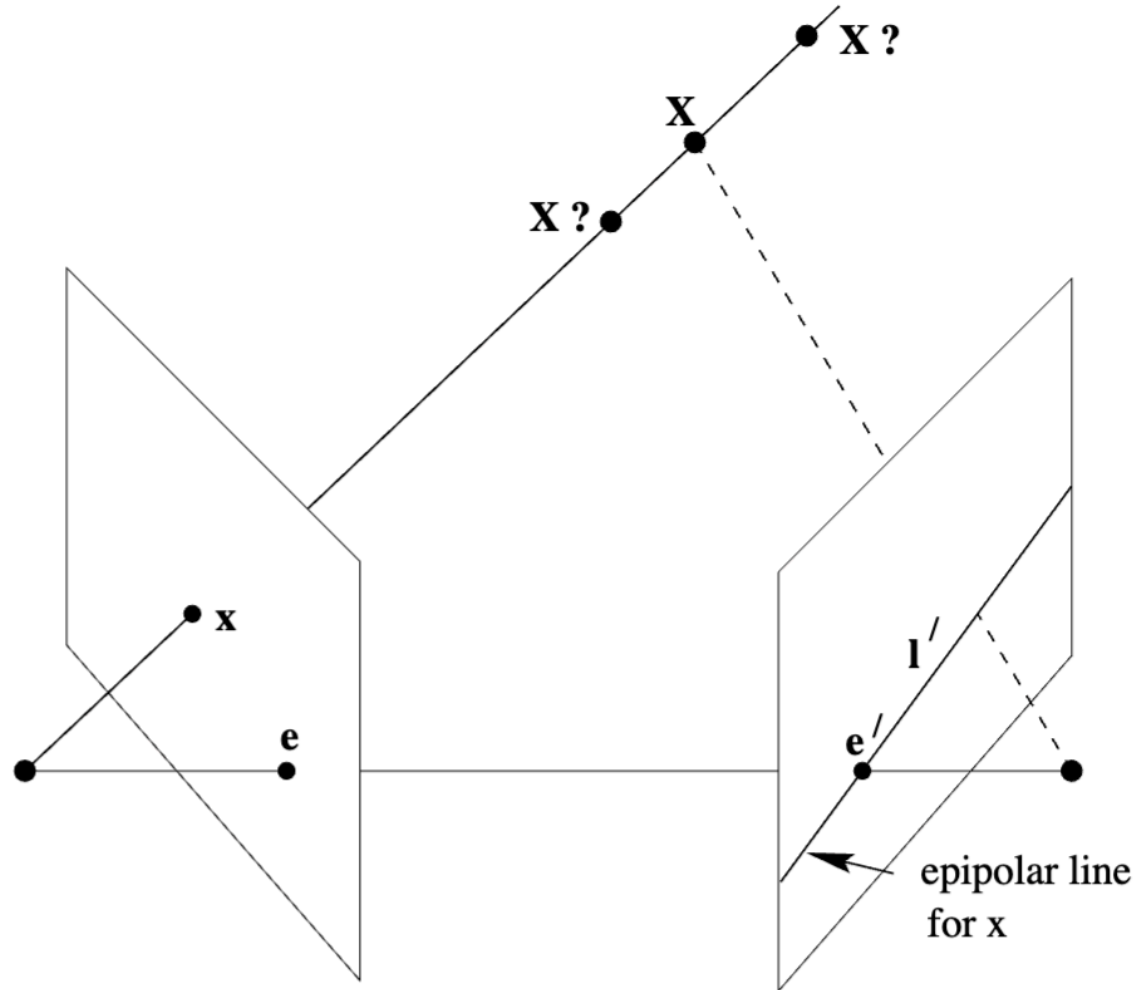
Equation $q^T E p = 0$ is a line equation in the p -plane with line coefficients $E^T q$. It is called the epipolar line in p -plane.

Epipolar Lines Constrain Point Correspondences!

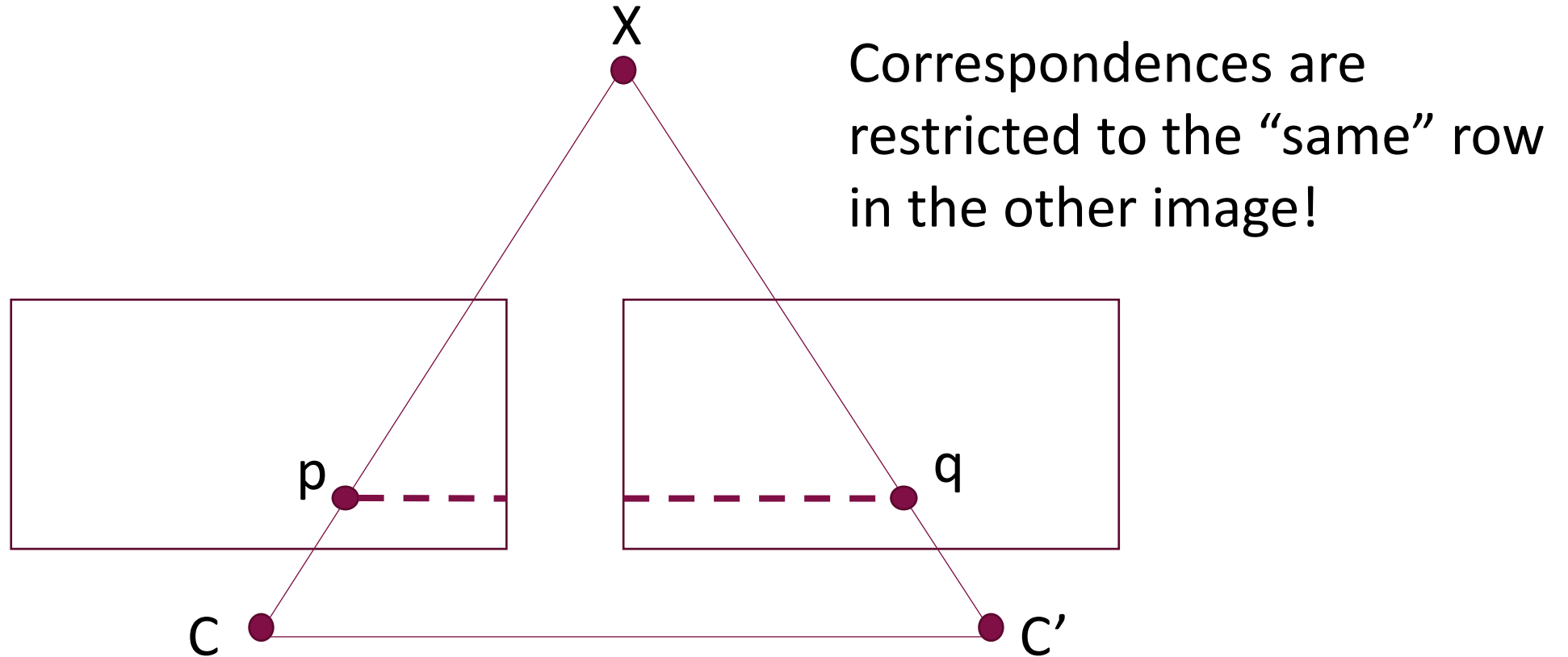


Given a point p = left image of a world point, the right image of that point is constrained to lie on the corresponding right epipolar line, and vice versa. (depending on depth of the world point).

Epipolar Lines Constrain Point Correspondences!



Special case: “frontoparallel” / “parallel stereo” cameras



Q: Where is the epipole = image of other camera center in each image?

A: Baseline is parallel to image plane(s) $\Rightarrow$ epipole is at infinity along x , and epipolar line is horizontal.

What if the cameras are “frontoparallel” / “parallel stereo”?



left image



right image

Longuet-Higgins' 8-Point Algorithm



Hugh Christopher Longuet-Higgins, quantum chemist turned cognitive scientist. 1981

Also see: [Hartley 1997: “In defense of the 8-point algorithm”](#)

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A computer algorithm for reconstructing a scene from two projections

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A simple algorithm for computing the three-dimensional structure of a scene from a correlated pair of perspective projections is described here, when the spatial relationship between the two projections is unknown. This problem is relevant not only to photographic surveying¹ but also to binocular vision², where the non-visual information supplied by the observer about the orientation and focal length of each eye is much less accurate than the optical information supplied by the retinal images themselves. The problem also arises in monocular perception of motion³, where the two projections represent views which are separated in time as well as space. As Marr and Poggio⁴ have noted, the fusing of two images to produce a three-dimensional percept involves two distinct processes: the establishment of a 1:1 correspondence between image points in the two views—the ‘correspondence problem’—and the use of the associated disparities for determining the distances of visible elements in the scene. I shall assume that the correspondence problem has been solved; the problem of reconstructing the scene then reduces to that of finding the relative orientation of the two viewpoints.

Photogrammetrists know that if a scene is photographed from two viewpoints, then the relationship between the camera positions is uniquely determined, in general, by the photographic coordinates of just five distinguishable points; but actually calculating the structure of the scene from five sets of image coordinates involves the iterative solution of five simultaneous third-order equations⁵. I show here that if the scene contains as many as eight points whose images can be located in each projection, then the relative orientation of the two projections, and the structure of the scene, can be computed, in general, from the eight sets of image coordinates by a direct method which

calls for nothing more difficult than the solution of a set of simultaneous linear equations.

Let P be a visible point in the scene, and let (X_1, X_2, X_3) and (X'_1, X'_2, X'_3) be its three-dimensional cartesian coordinates with respect to the two viewpoints. The ‘forward’ coordinates X_i and X'_i are necessarily positive. The image coordinates of P in the two views may then be defined as

$$\begin{aligned} (x_1, x_2) &= (X_1/X_3, X_2/X_3), \\ (x'_1, x'_2) &= (X'_1/X'_3, X'_2/X'_3) \end{aligned} \quad (1)$$

and it is convenient to supplement them with the dummy coordinates

$$x_3 = 1, \quad x'_3 = 1 \quad (2)$$

so that one can then write

$$x_\mu = X_\mu/X_3, \quad x'_\mu = X'_\mu/X'_3 \quad (\mu, \nu = 1, 2, 3) \quad (3)$$

As the two sets of three-dimensional coordinates are connected by an arbitrary displacement, we may write

$$X'_i = \mathbf{R}_{ij}(X_j - \mathbf{T}_i) \quad (4)$$

where $\mathbf{T}$ is an unknown translational vector and $\mathbf{R}$ is an unknown rigid rotation matrix. (In this and subsequent equations I sum over repeated Greek subscripts.) The rotation $\mathbf{R}$ satisfies the relationships

$$\mathbf{R}\mathbf{R}^T = \mathbf{I} = \mathbf{R}^T\mathbf{R}, \quad \det \mathbf{R} = 1 \quad (5)$$

and it is convenient to adopt the length of the vector $\mathbf{T}$ as the unit of distance:

$$\mathbf{T}_i^2 = \mathbf{T}_1^2 + \mathbf{T}_2^2 + \mathbf{T}_3^2 = 1 \quad (6)$$

I begin by establishing a general relationship between the two sets of image coordinates—a relationship which expresses the condition that corresponding rays through the two centres of projection must intersect in space. We define a new matrix $\mathbf{Q}$ by

$$\mathbf{Q} = \mathbf{R}\mathbf{S} \quad (7)$$

where $\mathbf{S}$ is the skew-symmetric matrix

$$\mathbf{S} = \begin{bmatrix} 0 & \mathbf{T}_3 & -\mathbf{T}_2 \\ -\mathbf{T}_3 & 0 & \mathbf{T}_1 \\ \mathbf{T}_2 & -\mathbf{T}_1 & 0 \end{bmatrix} \quad (8)$$

Equation (8) may be written as

$$S_{\lambda\nu} = \epsilon_{\lambda\nu\sigma} \mathbf{T}_\sigma \quad (9)$$

where $\epsilon_{\lambda\nu\sigma} = 0$ unless (λ, ν, σ) is a permutation of $(1, 2, 3)$, in which case $\epsilon_{\lambda\nu\sigma} = \pm 1$ depending on whether this permutation is even or odd. It follows from equations (4)–(9) that

$$\begin{aligned} X'_i \mathbf{Q}_{\mu\nu} X_\nu &= \mathbf{R}_{ij}(X_j - \mathbf{T}_i) \mathbf{R}_{k\ell} \epsilon_{\lambda\nu\sigma} \mathbf{T}_\sigma X_\nu \\ &= (X_i - \mathbf{T}_i) \epsilon_{i\mu\nu} \mathbf{T}_\nu X_\nu \end{aligned} \quad (10)$$

but because the quantity $\epsilon_{i\mu\nu}$ is antisymmetric in every pair of its subscripts, the right-hand side vanishes identically:

$$X'_i \mathbf{Q}_{\mu\nu} X_\nu = 0 \quad (11)$$

Dividing equation (11) by $X'_i X_3$ we arrive at the desired relationship between the image coordinates:

$$x'_i \mathbf{Q}_{\mu\nu} x_\nu = 0 \quad (12)$$

The next step is to determine the nine elements $\mathbf{Q}_{\mu\nu}$. There will be one equation of type (12) for every point P_i , namely

$$(x'_{i\mu} x_{i\nu}) \mathbf{Q}_{\mu\nu} = 0 \quad (13)$$

and in this equation the nine quantities $(x'_{i\mu} x_{i\nu})$ are presumed to be known. The ratios of the nine unknowns $\mathbf{Q}_{\mu\nu}$ can therefore be obtained, in general, by solving eight simultaneous linear equations of type (13), one for each of eight visible points $P_1, \dots, P_8$. I shall not yet discuss the special circumstances under which the solution fails; for the present merely note that if the eight equations (13) are independent, their solution is entirely straightforward from a computational point of view.

8-Point Algorithm

- Recall that each correspondence gives us one linear equation in the unknowns E

Is this really linear in E ?

$q_i^T E_{3 \times 3} p_i = 0$ is a single equation that is linear in the elements of E
Can write this out explicitly as below.

If

$$E = (e_1 \ e_2 \ e_3)$$

then epipolar constraint can be rewritten as

$$q^T (e_1 \ e_2 \ e_3) \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = q^T (p_x e_1 + p_y e_2 + p_z e_3)$$

$$= \begin{pmatrix} p_x q^T & p_y q^T & p_z q^T \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = 0$$

“ $a_{1 \times 9}$ ”

This equation is linear

“ $E'_{9 \times 1}$ ”

8-Point Algorithm

Let $\vec{a} = (p_x q^T \quad p_y q^T \quad p_z q^T)$

$$\begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{pmatrix}_{n \times 9} E' = 0$$

One row per point correspondence

where a_i is the known 1×9 vector of image points and E' is the essential matrix re-organized into a 9×1 column vector.

E' has to be in the null-space of $\begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{pmatrix}$.

Does this remind you of something?
Hint: 4-Point Collineations, PnP, ...

Solution: set E' to the last right singular vector of $A_{n \times 9}$

Rank of A in the 8-Point Algorithm

Problem: What if we have >8 points and $A_{n \times 9}$ is full-rank (i.e. rank 9)?

Not a problem! The last singular vector of A already solves for:

$$\operatorname{argmin}_{E': ||E'||_2=1} ||AE'||_2,$$

which is the closest thing to finding a null vector for a non-singular matrix

Problem: A may be ***too low-rank***. For example, with perfect measurements, A is rank 8 only if points do not all lie on any “quadric” surface*, including planes, cylinders, ellipsoids etc. Otherwise, $\operatorname{rank}(A) < 8$! This can happen quite frequently in practice, e.g., smartphone moving facing a wall.

No real solution. Avoid relying on correspondences exclusively from such surfaces to the extent possible. If still $\operatorname{rank} < 8$, give up on that pair of views.

*<https://tutorial.math.lamar.edu/classes/calci/quadricsurfaces.aspx>

After solving for E , not Quite Done Yet!

$E = \hat{T}R$ has fewer than 8 DOF. T has 3 DOF (+3), R has 3 DOF (+3), and E is scale invariant (-1), so total 5 DOF. **So not all 3x3 matrix is a valid essential matrix.**

- **Problem:** Given the above, how to ensure that the estimated E is a valid essential matrix?
- **Problem:** How to decompose E into the $\hat{T}, R$ required in SfM?

We will revisit these problems soon!

Note: Do we really need 8 Points to solve SfM?

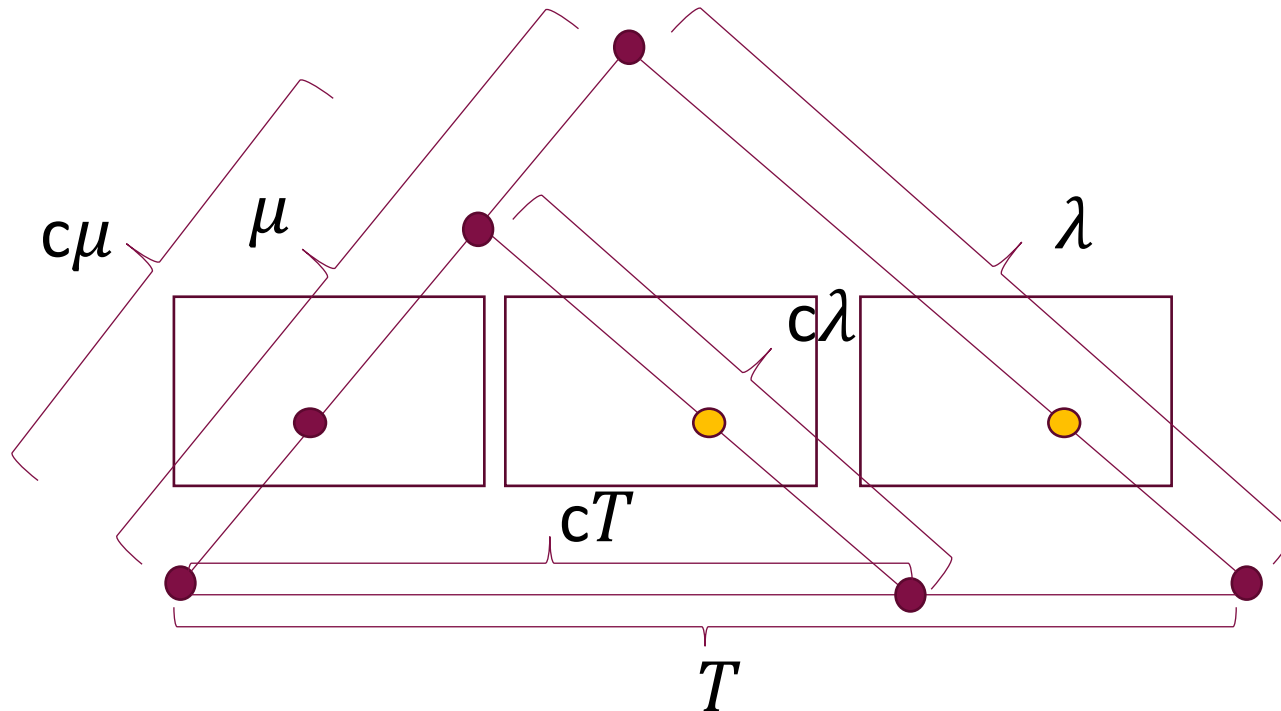
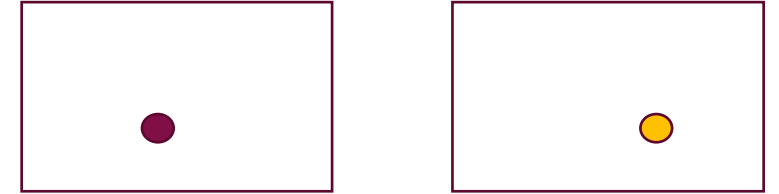
- Actually, there are methods for SfM with as few as 5 point correspondences, corresponding to 5 degrees of freedom in R, T (minus scale ambiguity)
- But much like our P3P Step 1, these involve solving complicated systems of higher degree equations. Instead, the 8-point algorithm offers a “direct” solution by solving a linear system of equations, and is widely used due to its convenience. (Much like the “direct solution” we saw for PnP, which needed $6 > 3$ 2D-3D point correspondences for PnP)

Note: SfM Scale Ambiguity Illustration

Remember, all you are given is calibrated coordinate correspondences between 2 image planes.

So, suppose one solution to SfM is R, T, λ, μ .

Then, an equally valid solution is $R, cT, c\lambda, c\mu$!



Scale ambiguity applies to purely visual SfM. If you have access to an accelerometer on a moving camera, can resolve scale ambiguity! (“Visual inertial odometry”).

CIS 5800

Machine Perception

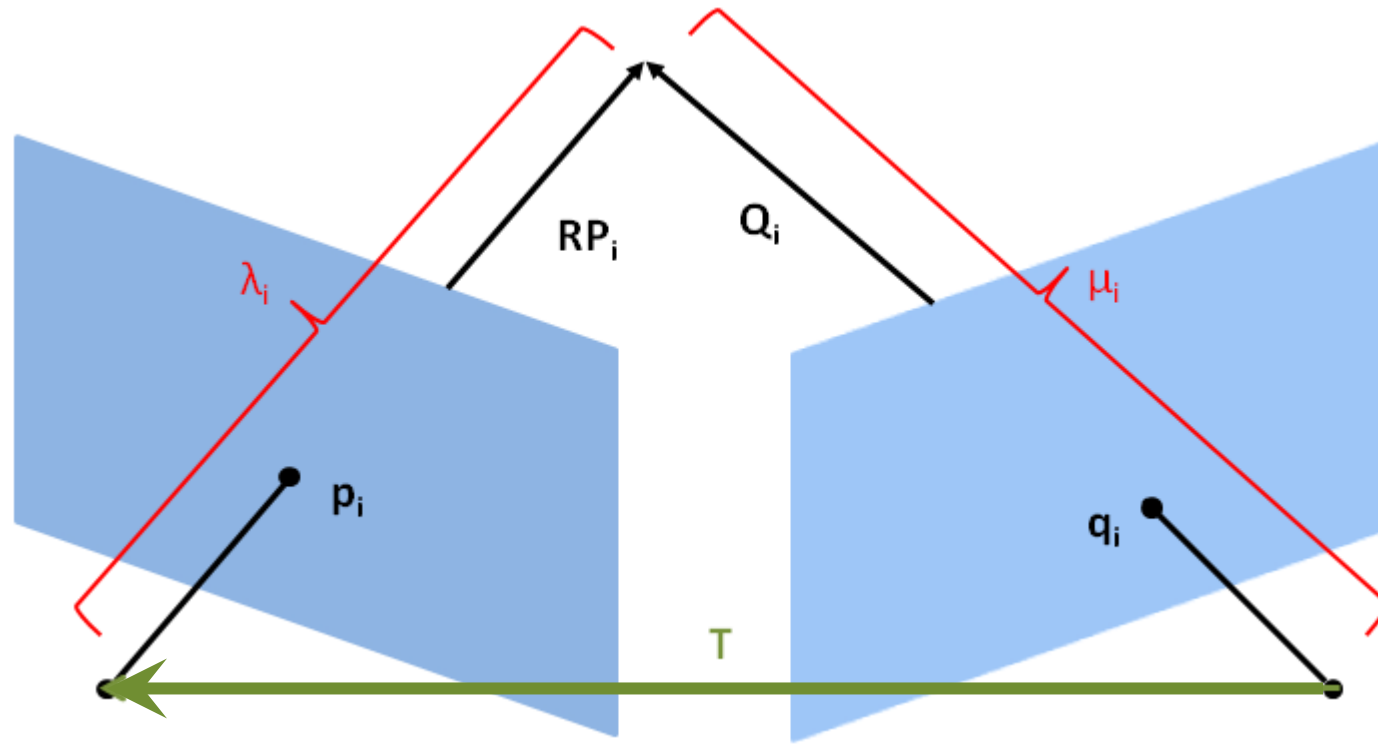
Instructor: Lingjie Liu

Lec 12: March 3, 2026

Administrivia

- Midterm exam coming up
 - Date reminder: Thursday March 26 (class hours, we'll 3:30pm exactly)
 - Syllabus: All material covered in class until Tue March 17.
 - Review questions pdf will be released this week
 - If you are unable to attend the midterm exam in person on March 26, please complete the form by March 17:
<https://docs.google.com/forms/d/e/1FAIpQLSfzXSMvI4DqEdeos-WDEMFbgn7I1BJJ-kpSGLVkolaJZgJsWg/viewform?usp=dialog>
 - Also, you need to contact the Weingarten Office for academic accommodations and send me the paperwork or approval from the Weingarten Office.

Recap: Two Calibrated Views of the Same 3D Scene

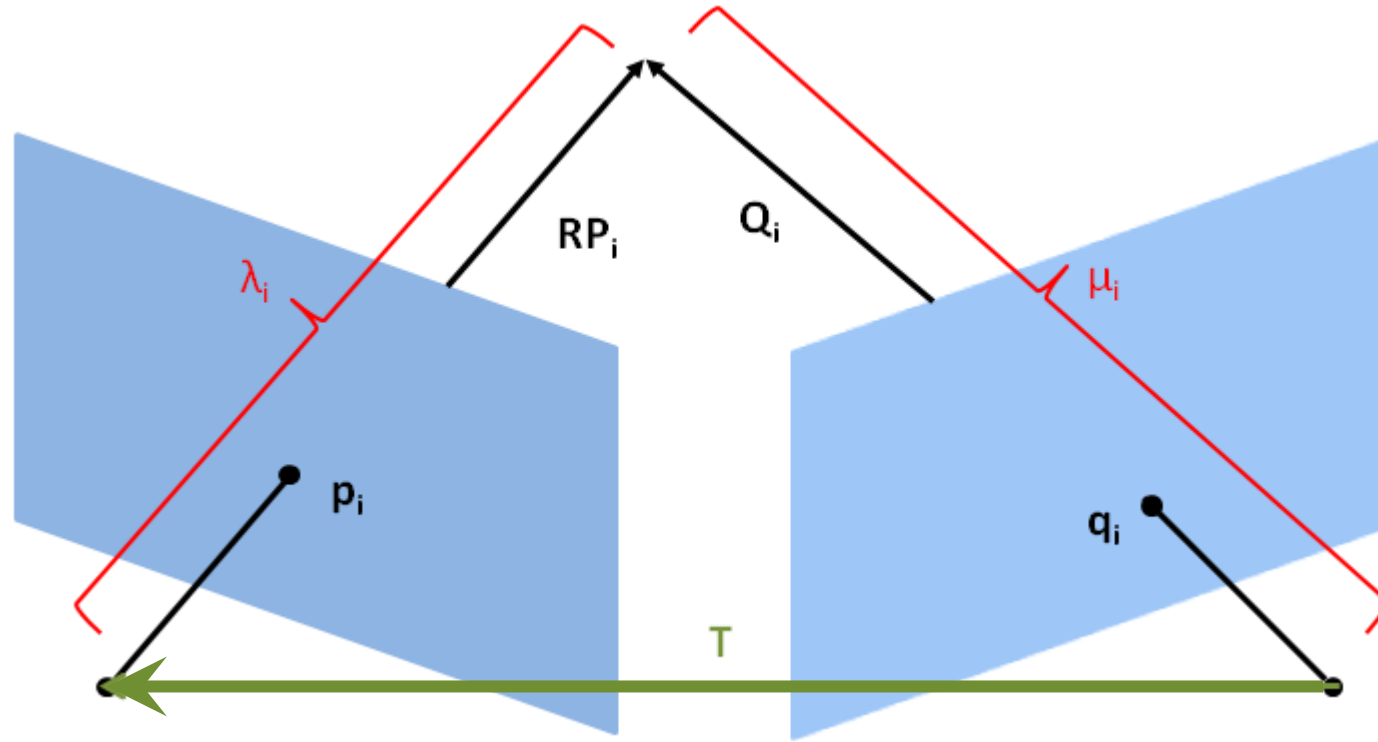


$$R(\lambda p) + T = \mu q$$

Given 2D correspondences (p, q)

Find motion R, T and depths λ, μ .

Recap: “Epipolar Constraints” Between Two Views of a Scene



We can eliminate the depths from $R(\lambda p) + T = \mu q$ and obtain the epipolar constraint:

$$q_i^T (T \times R p_i) = 0$$

Recap: The Essential Matrix E

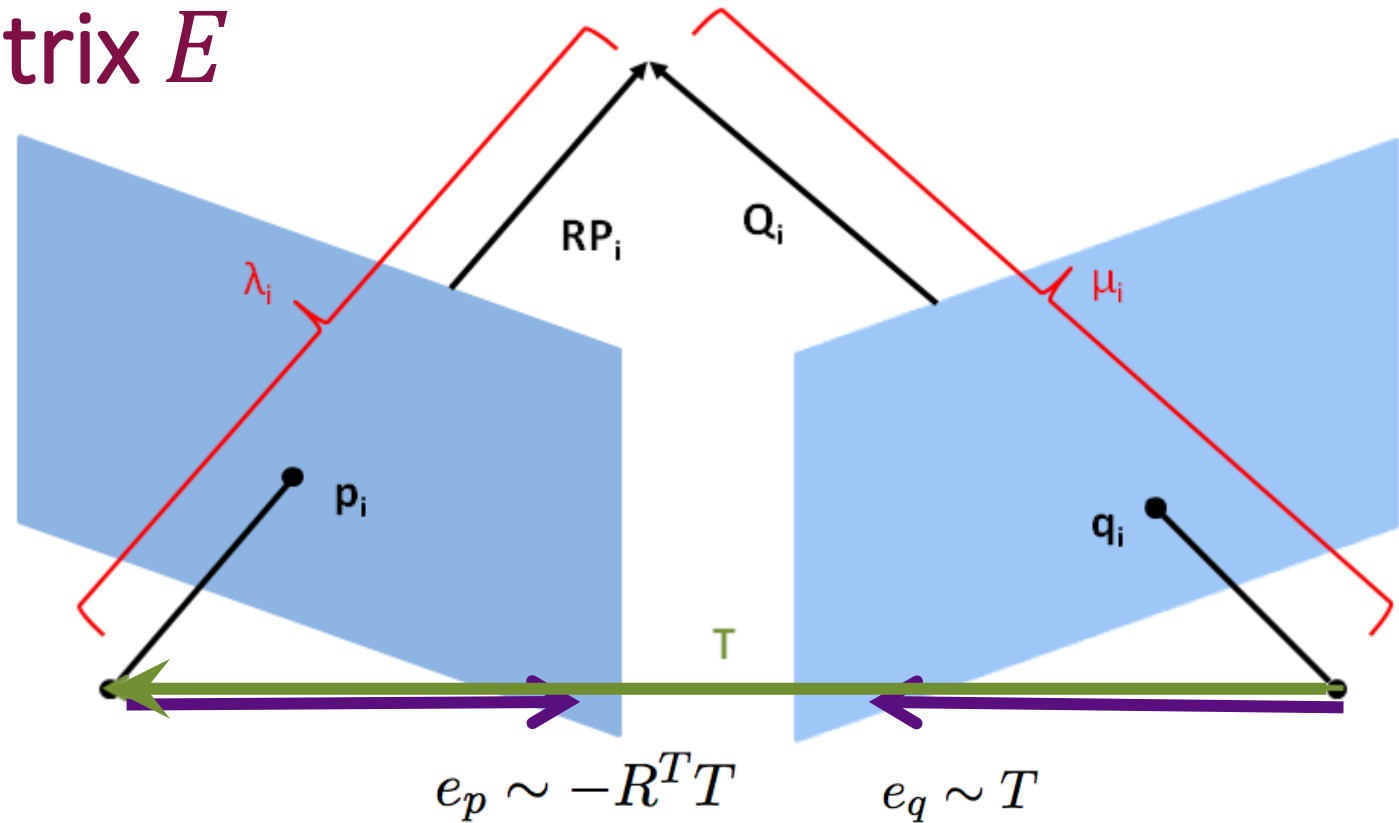
We had: $\mathbf{q}_i^T (\mathbf{T} \times \mathbf{R} \mathbf{p}_i) = 0$

$$\Rightarrow \mathbf{q}_i^T (\hat{\mathbf{T}} \mathbf{R}) \mathbf{p}_i = 0$$

Renaming $\mathbf{E} = (\hat{\mathbf{T}} \mathbf{R})$:

$$\mathbf{q}_i^T \mathbf{E} \mathbf{p}_i = 0$$

↓
“Essential matrix”



Now linear in the new unknowns $\mathbf{E}_{3 \times 3}$! But will need to recover $\mathbf{T}_{3 \times 1}$, $\mathbf{R}_{3 \times 3}$ later.

Recap: 8-Point Algorithm

- Recall that each correspondence gives us one linear equation in the unknowns E

Is this really linear in E ?

$q_i^T E_{3 \times 3} p_i = 0$ is a single equation that is linear in the elements of E
Can write this out explicitly as below.

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“ $a_{1 \times 9}$ ”

This equation is linear

“ $E'_{9 \times 1}$ ”

Recap: 8-Point Algorithm

Let $\vec{a} = (p_x q^T \quad p_y q^T \quad p_z q^T)$

$$\begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{pmatrix} E' = 0$$

One row per point correspondence

$$n \times 9$$

where a_i is the known 1×9 vector of image points and E' is the essential matrix re-organized into a 9×1 column vector.

E' has to be in the null-space of $\begin{pmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{pmatrix}$.

Does this remind you of something?
Hint: 4-Point Collineations, PnP, ...

Solution: As before, set E' to the last right singular vector of $A_{n \times 9}$

Recap: After solving for E , not Quite Done Yet!

$E = \hat{T}R$ has fewer than 8 DOF. T has 3 DOF (+3), R has 3 DOF (+3), and E is scale invariant (-1), so total 5 DOF. **So not any 3x3 matrix is a valid essential matrix.**

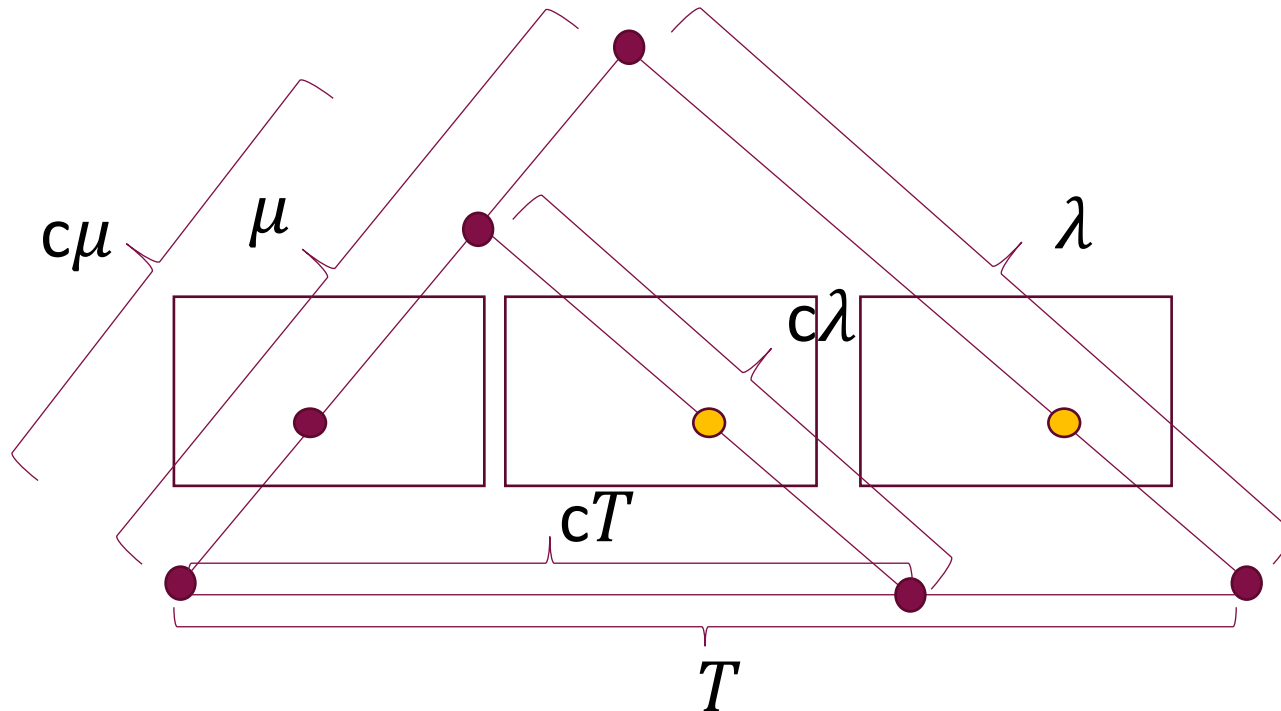
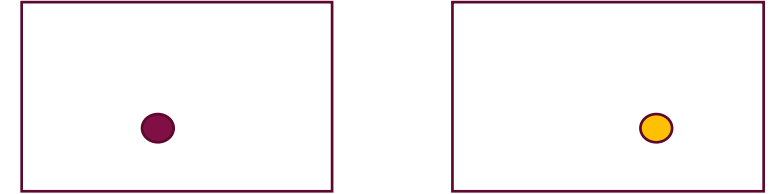
- **Problem:** Given the above, how to ensure that the estimated E is a valid essential matrix?
- **Problem:** How to decompose E into the $\hat{T}, R$ required in SfM?

Note: SfM Scale Ambiguity Illustration

Remember, all you are given is calibrated coordinate correspondences between 2 image planes.

So, suppose one solution to SfM is R, T, λ, μ .

Then, an equally valid solution is $R, cT, c\lambda, c\mu$!



Scale ambiguity applies to purely visual SfM. If you have access to an accelerometer on a moving camera, can resolve scale ambiguity! (“Visual inertial odometry”).

SfM Scale Ambiguity and Miniature Movie Sets!



Grand “establishing shots” in movies are often filmed with miniature sets (and green screens for background)

No one can tell, because of scale ambiguity!



Fundamental matrix

- Essential matrix E connects image plane coordinates i.e. “calibrated coordinates” for points $p = K_p^{-1}(u_p, v_p, 1)^T$ and $q = K_q^{-1}(u_q, v_q, 1)^T$.
- Fundamental matrix F assumes that we are still in “pixel land”
- So:

$$q^T E p = 0 \quad \text{can be written as:} \quad (K_q q)^T F (K_p p) = 0$$

Recap: After solving for E , not Quite Done Yet!

$E = \hat{T}R$ has fewer than 8 DOF. T has 3 DOF (+3), R has 3 DOF (+3), and E is scale invariant (-1), so total 5 DOF. **So not any 3x3 matrix is a valid essential matrix.**

- **Problem:** Given the above, how to ensure that the estimated E is a valid essential matrix?
- **Problem:** How to decompose E into the $\hat{T}, R$ required in SfM?

Review:

- Singular Value Decomposition (SVD)
- Eigenvalues

Constructing Valid Essential Matrices and Decomposing Them

Necessary and sufficient condition: E is essential iff $\sigma_1(E) = \sigma_2(E) \neq 0$ and $\sigma_3(E) = 0$.

Part 1: Proving ‘necessary’ (“If E is essential, then ...”) will tell us about properties of essential matrices, so we can correct the E matrices from the direct method to become valid.

Part 2: Proving ‘sufficient’ (“If singular values ..., then ...”) will help us solve R, T from E for a particular pair of cameras.

Proof Sketch Part 1: Singular Values of A Valid Essential Matrix

$$\sigma_1(E) = \sigma_2(E) \neq 0 \text{ and } \sigma_3(E) = 0.$$

a square matrix whose determinant is zero

E is a singular matrix (Because $E = \hat{T}R$, and $\det(\hat{T}) = 0$)

So smallest singular value is indeed zero!

$$\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

For the two other singular values of E , recall that they are square roots of the eigenvalues of EE^T .

Finding eigenvalues $\Rightarrow$ setting $\det(EE^T - \lambda I) = 0$ i.e. “characteristic polynomial”

$$E = \hat{T}R, \Rightarrow E^T = R^T \hat{T}^T$$

$$EE^T = \hat{T}\hat{T}^T \\ = - \begin{bmatrix} t_x^2 & t_x t_y & t_x t_z \\ t_x t_y & t_y^2 & t_y t_z \\ t_x t_z & t_y t_z & t_z^2 \end{bmatrix} + \|T\|^2 I$$

Proof Sketch Part 1: Singular Values of A Valid Essential Matrix

$$\sigma_1(E) = \sigma_2(E) \neq 0 \text{ and } \sigma_3(E) = 0.$$

$$EE^T = \hat{T}\hat{T}^T \\ = - \begin{bmatrix} t_x^2 & t_x t_y & t_x t_z \\ t_x t_y & t_y^2 & t_y t_z \\ t_x t_z & t_y t_z & t_z^2 \end{bmatrix} + \|T\|^2 I$$

If we solve the characteristic polynomial $\det(EE^T - \lambda I) = 0$ we will find two eigenvalues both equal to $\|T\|^2$. (Bonus exercise: try this out by hand)

$$\text{So, } \sigma_1(E) = \sigma_2(E) = \|T\| \text{ and } \sigma_3(E) = 0$$

Side note: the ‘third singular vector’ of E (null vector because $\sigma_3 = 0$) is nothing but the translation vector T because:

$$EE^T T = \hat{T}\hat{T}^T T = T \times (-T \times T) = 0!$$

So we already know T in terms of E ! (the 3rd left singular vector of E)

Proof Sketch Part 1: Singular Values of A Valid Essential Matrix

Hence, we have proved that **if a matrix is essential, namely, can be decomposed as the product of an antisymmetric $\hat{T}$ and a special orthogonal R then its singular values are $\sigma_1 = \sigma_2 > 0$ and $\sigma_3 = 0$.**

$$\sigma_1(E) = \sigma_2(E) \neq 0 \text{ and } \sigma_3(E) = 0.$$

Condition A

Utility: Having obtained an initial estimate E through direct solution of ≥ 8 epipolar constraints, we may enforce “condition A” above on it, by:

1. Compute SVD $E = U \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix} V^T$

2. Then, set $E_{new} = U \begin{bmatrix} (\sigma_1 + \sigma_2)/2 & & \\ & (\sigma_1 + \sigma_2)/2 & \\ & & 0 \end{bmatrix} V^T$

This satisfies
 $\operatorname{argmin}_{E_{new}} ||E_{new} - E||_F^2$ s.t.
 E_{new} meets “condition A”.

Since E is scale-invariant,
optionally, can also just
set Σ to $\operatorname{diag}(1,1,0)$

We don't yet know how to get R, T from E

Proof Part 2: Construction of E as $\hat{T}R$

Necessary and sufficient condition: E is essential iff $\sigma_1(E) = \sigma_2(E) \neq 0$ and $\sigma_3(E) = 0$.

We have to prove the sufficient condition:

If the singular values of a matrix are $\sigma_1 = \sigma_2 > 0$ and $\sigma_3 = 0$ then the matrix can be decomposed into the product of an antisymmetric $\hat{T}$ and a special orthogonal R .

i.e. valid rotation matrix R , orthonormal with determinant +1 (right-handed coordinate system)

Proof Part 2: Construction of E as $\hat{T}R$

$$\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

Consider the simplest matrix satisfying $\sigma_1 = \sigma_2$ and $\sigma_3 = 0$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

It indeed can be decomposed into antisymmetric / skew and rotation, e.g.:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

#1

skew-symmetric $\hat{T}_{-z}$ $\times$ rotation $R_{z,\pi/2}$

$$T_{-z} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

and rotation by 90° about z axis

Proof Part 2: Construction of E as $\hat{T}R$

If the singular values of a matrix are $\sigma_1 = \sigma_2 > 0$ and $\sigma_3 = 0$ then the matrix can be decomposed into the product of an antisymmetric $\hat{T}$ and a special orthogonal R .

We need one more piece before we can prove this.

If Q is orthogonal ($Q^T Q = I$), then

$$\widehat{Qa} = Q\hat{a}Q^T$$

#2

Proof: $\widehat{Qa}b = Qa \times b = Q(a \times Q^T b) = Q\hat{a}Q^T b.$

It doesn't matter whether you rotate first and then take cross-product or the other way around. i.e., For a rotation R , $R(\mathbf{a} \times \mathbf{b}) = R\mathbf{a} \times R\mathbf{b}$

Now let's put #1 and #2 together

Proof Part 2: Construction of E as $\hat{T}R$

For a general essential matrix E that has two equal singular values and one singular value 0, we can write:

$$E = U \begin{bmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

$$= \sigma U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

$$= \sigma U \hat{T}_{-z} R_{z,\pi/2} V^T$$

$$= \sigma U \hat{T}_{-z} U^T U R_{z,\pi/2} V^T$$

$$= \sigma (\widehat{UT_{-z}}) (UR_{z,\pi/2} V^T)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

skew-symmetric $\hat{T}_{-z}$ $\times$ rotation $R_{z,\pi/2}$

#1

$$U^T U = I$$

If Q is orthogonal ($Q^T Q = I$), then

$$\widehat{Qa} = Q\hat{a}Q^T$$

#2

Proof Part 2: Construction of E as $\hat{T}R$

For a general essential matrix E , we can write it as the product of a skew-symmetric matrix, and an orthogonal matrix:

$$E = \sigma(\widehat{UT_{-z}}) (UR_{z, \pi/2}V^T)$$

Skew symmetric; it represents cross product with UT_{-z} = the last column of U = last left singular vector of E (w. flipped sign)

T = last left singular vector of E (w. flipped sign*)

orthogonal, since product of orthogonal matrices

R

*We'll soon see the sign flipping isn't a big deal

Q: Is this the *only* way to decompose $E = \hat{T}R$?

No!

Uniqueness of SfM solution?

$$E = \sigma(\widehat{UT_{-z}}) (UR_{z, \pi/2} V^T)$$

We used the following decomposition:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

skew-symmetric $\widehat{T_{-z}}$ x rotation $R_{z, +\pi/2}$

But what if we had instead used the equally valid:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

skew-symmetric $\widehat{T_{+z}}$ x rotation $R_{z, -\pi/2}$

We would have ended up with a different estimate of $\hat{T}, R$:

- $\hat{T} = \sigma(\widehat{UT_{+z}})$
- $R = UR_{z, -\pi/2} V^T$

Uniqueness of SfM solution?

- So, for a given E matrix, there are two possible decompositions $\hat{T}, R$
- But this is not all. If E is a valid solution to $q^T E p = 0$, then so is $-E$!
 - And $-E$ induces its own two decompositions into $\hat{T}, R$:
 - $E = \sigma(\widehat{UT_{-z}})(UR_{z,+\pi/2}V^T) \Rightarrow -E = \sigma(\widehat{UT_{+z}})(UR_{z,+\pi/2}V^T)$, and
 - $E = \sigma(\widehat{UT_{+z}})(UR_{z,-\pi/2}V^T) \Rightarrow -E = \sigma(\widehat{UT_{-z}})(UR_{z,-\pi/2}V^T)$

The Four Possible R, T decompositions of $\pm E$

If $E = U\Sigma V^T = U \begin{bmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$, there are four solutions for the pair $(\hat{T}, R)$

$$\pm E = \begin{matrix} \hat{T} & R \\ \sigma(\widehat{UT}_{-z}) & (UR_{z,+\pi/2}V^T) \\ \sigma(\widehat{UT}_{+z}) & (UR_{z,+\pi/2}V^T) \\ \sigma(\widehat{UT}_{+z}) & (UR_{z,-\pi/2}V^T) \\ \sigma(\widehat{UT}_{-z}) & (UR_{z,-\pi/2}V^T) \end{matrix}$$

Can get rid of the scale σ ,
no harm done.

Last left singular vector of E , with
either + or - sign

Can disambiguate by enforcing positive
depths for all points (most points if
noisy) in both cameras.

But how to get depths?

Note: Both R
matrices are not
guaranteed to have
determinant +1.
Could be -1. But at
least one with +1.

Computing depths λ_i, μ_i through “triangulation”

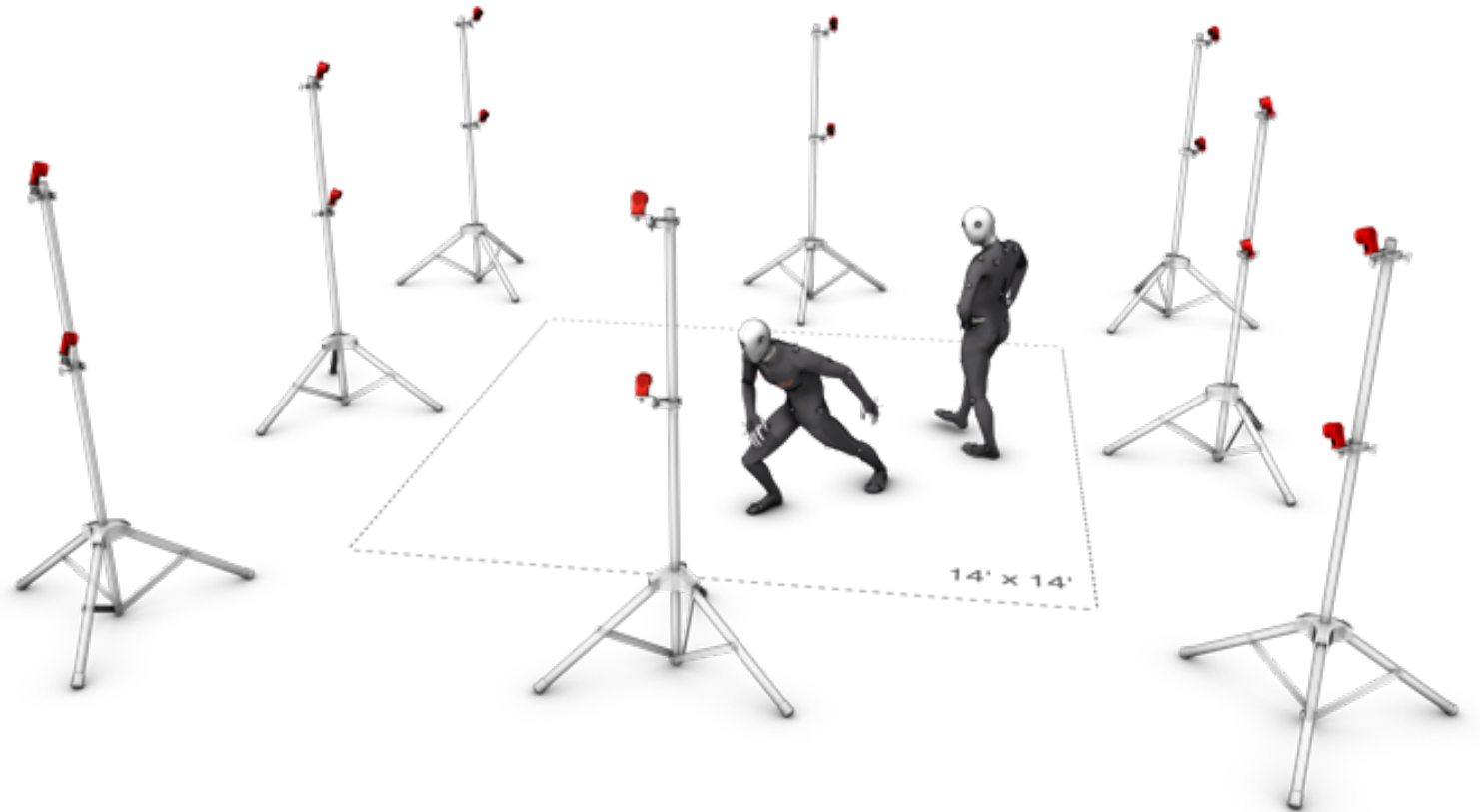
Triangulation is possible if we have computed R and T but again up to a scale factor. Set $\|T\| = 1$:

$$\underbrace{(q_i - Rp_i)}_{3 \times 2} \underbrace{\begin{pmatrix} \mu_i \\ \lambda_i \end{pmatrix}}_{2 \times 1} = \underbrace{T}_{3 \times 1}$$

There are then 3 equations with 2 unknowns λ_i and μ_i for each point.

Solve with pseudo-inverse.

Triangulation is independently useful! E.g. MoCap



The full two-view 8-point algorithm

Direct solution of E

- 1 Build the homogeneous linear system by stacking epipolar constraints $q_i^T (T \times Rp_i) = 0, i = 1, \dots, 8$:

$$\begin{bmatrix} \vdots \\ (q_i \otimes p_i)^T \\ \vdots \end{bmatrix} \begin{bmatrix} e'_1 \\ e'_2 \\ e'_3 \end{bmatrix}$$

$A \ (8 \times 9)$

- 2 Let $\begin{bmatrix} e'_1 \\ e'_2 \\ e'_3 \end{bmatrix}$ be the nullspace of A (if $\sigma_8 \approx 0$ give up)

The full two-view 8-point algorithm

Decompose
Into T, R

Make E valid

- ③ $\begin{bmatrix} e'_1 & e'_2 & e'_3 \end{bmatrix} = U \text{diag} (\sigma'_1 \ \sigma'_2 \ \sigma'_3) V^T$. Then use the following estimate of the essential matrix:

$$E = U \text{diag} \left(\frac{\sigma'_1 + \sigma'_2}{2}, \frac{\sigma'_1 + \sigma'_2}{2}, 0 \right) V^T$$

- ④ $T = \pm \hat{u}_3 \quad R = UR_{Z,\pi/2}V^T$ or $R = UR_{Z,-\pi/2}V^T$
- ⑤ Try all four pairs (T, R) to check if reconstructed points are **in front** of the cameras $\boxed{\lambda q = \mu Rp + T}$ give $\lambda, \mu > 0$.

CIS 5800

Machine Perception

Instructor: Lingjie Liu

Lec 13: March 5, 2026

Administrivia

- Next week is Spring Break
- Deadlines of HW 2 and HW3 are March 17 and March 18 respectively
- Midterm Exam (March 26)
 - The format is slightly different from that of the review questions, but the coverage of the key knowledge points will be the same.
 - 5 long questions (50pts in total = 10pts x 5)
 - Knowledge Points: Everything covered up to the class of March 17.
- Combined slides for your mid-term exam preparation will be uploaded to Canvas tomorrow.
- Review for Knowledge Points: Next Class (March 24).
- We will have additional OHs in the week after Spring Break. TAs will announce the specific times asap.

2-View SfM: How we recovered $R, T, \{\lambda_i, \mu_i\}_{i=1}^n$

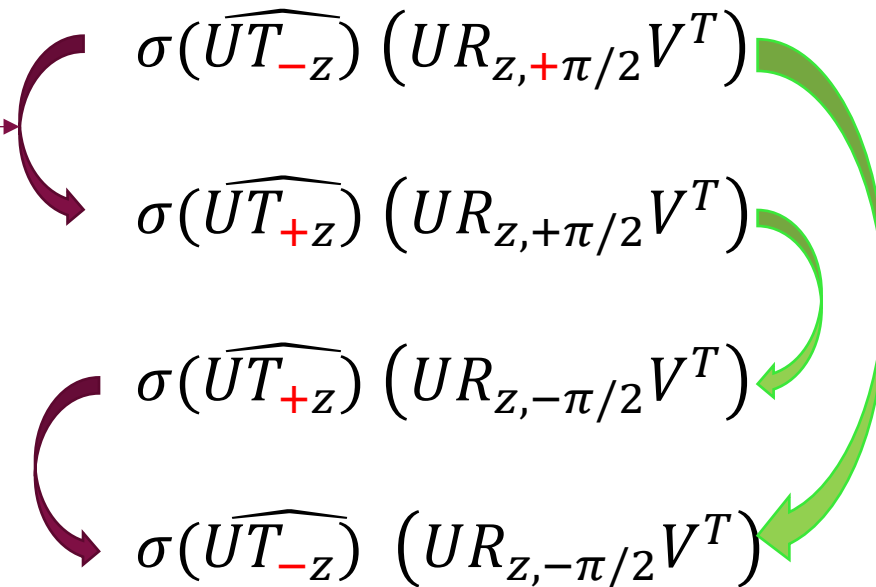
- Invent an object $E = \hat{T}R$
- Express epipolar constraint for every point correspondence $q_i^T E p_i = 0$
- Solve equations to get initial estimate E_0
- $SVD(E_0) = U\Sigma V^T$, Then set $E = U\text{diag}(1, 1, 0)V^T$
- Then, long proof to show you can set $\pm$ (last left singular vector of E) to be T , and there are two possible rotation matrices you can pair each translation vector with.
 - Solve depth by solving $\mu_i R p_i + T = \lambda_i q_i$ for each of these 4 solutions.
 - Settle on the solution that has all (or most) depths positive.
(Sidenote/Q: Are λ_i and μ_i really “depths?”)

What is the meaning of these ambiguities in the solution to SfM?

Types of Ambiguity in 2-View SfM Solutions

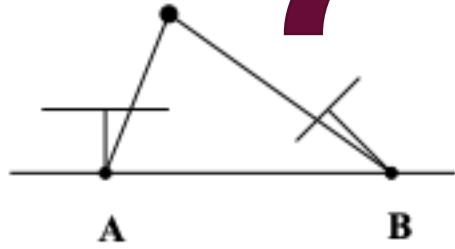
Mirror ambiguity: If T is a solution, then $-T$ is a solution, too. There is no way to disambiguate from the epipolar constraint: $q^T(-T \times Rp) = 0$.

Just extending scale ambiguity to negative scales. Some people don't even count this as a "true" ambiguity of SfM.

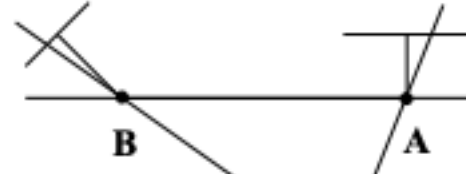


Twisted pair ambiguity: If R is a solution, then also $R_{T,\pi}R$ is a solution. The first camera is "twisted" around the baseline 180 degrees.

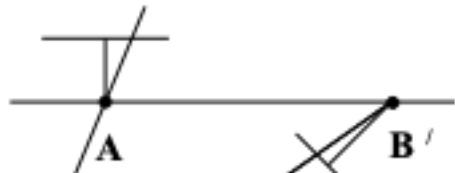
Mirror ambiguity



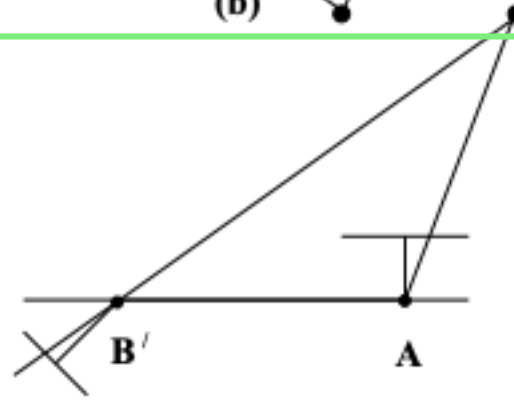
(a)



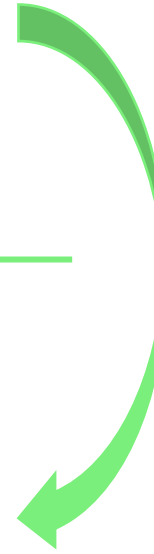
(b)



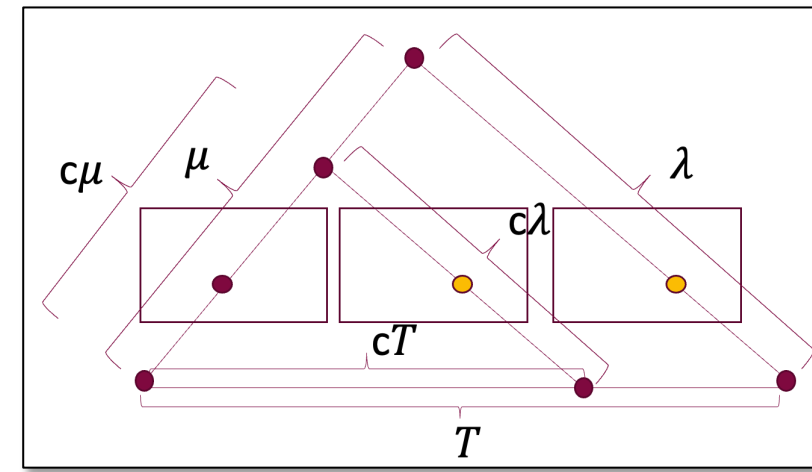
(c)



(d)



Twisted pair ambiguity



What *is* “solving” when solutions have “ambiguity”?

- We said we could solve for camera pose given 3 2D->3D correspondences (P3P)
 - But there was ambiguity: 4 solutions!

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P3P Step 1: The algebraic drudgery

Reduces to two quadratic equations in u and v .

$$d_{13}^2(u^2 + v^2 - 2uv \cos \delta_{23}) = d_{23}^2(1 + v^2 - 2v \cos \delta_{13}) \quad (1)$$

$$d_{12}^2(1 + v^2 - 2v \cos \delta_{13}) = d_{13}^2(u^2 + 1 - 2u \cos \delta_{12}) \quad (2)$$

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- Solve Eqn (1) for u^2 in terms of u, v, v^2 (and constants).
- Plug this solution into Eqn (2), so that it has no u^2 term. Solve for u in terms of v, v^2 , and constants.
- Plug this solution for u back into Eqn (1), so that it has no more u or u^2 . Instead, it is a 4th degree equation in v . *Get the 4 real solutions analytically.*
- Then plug back into the solution found in step b) above, to get u .
- Then get d_1 from the quadratic equations on the last page.
- Then plug back into $d_2 = ud_1$ and $d_3 = vd_1$ from the last page to get d_2, d_3 .

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